

Singular stochastic PDEs*

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1 Introduction

Stochastic partial differential equations (SPDEs) naturally arise in the mathematical description of systems influenced by randomness, such as fluctuating interfaces, turbulent fluids, or fields in statistical mechanics and quantum field theory. While many SPDEs can be analysed using classical techniques, a particularly challenging class—singular SPDEs—has become a central topic in modern probability theory and mathematical physics. These equations are singular because their solutions are too irregular for the nonlinear terms to be well-defined in the classical sense. This phenomenon parallels the ultraviolet divergences in quantum field theory (QFT), where renormalization is needed to define interacting models.

The first part of the course will introduce and motivate singular SPDEs, explain the origin of their singularities, and discuss the necessity of renormalization in constructing

*Under construction — please let me know about errors and typos.

their solutions. We will review core examples, including the KPZ and stochastic quantization equations, emphasizing their links to quantum field theory and to scaling limits of microscopic models.

The second part will focus on the flow equation approach, a robust framework applicable to a broad class of singular SPDEs—including those with fractional Laplacians—throughout the entire subcritical regime. Inspired by Wilson’s renormalization group, this method studies the coarse-grained process, which captures the behaviour of solutions across spatial scales. The corresponding flow equation describes how the nonlinear terms in the effective dynamics evolve with the coarse-graining scale, playing a role analogous to the Polchinski equation in QFT. The renormalization problem is then solved inductively by imposing appropriate boundary conditions on the flow equation.

2 White noise and stochastic heat equation

The SPDEs we will consider are driven by white noise (or its periodic analogue). Formally, white noise ξ on $\mathbb{R}^n$ is described as a centred Gaussian field with covariance

$$\mathbb{E} \xi(x) \xi(y) = \delta(x - y),$$

where δ denotes the n -dimensional Dirac delta distribution.

This identity already indicates that $\xi(x)$ cannot be interpreted as an ordinary random variable for fixed $x \in \mathbb{R}^n$. Indeed, there is no pointwise-defined coordinate process. Instead, white noise is rigorously defined as a random distribution, that is, as a random element of $\mathcal{S}'(\mathbb{R}^n)$, the space of tempered distributions. Concretely, for every test function $f \in \mathcal{S}(\mathbb{R}^n)$, the pairing $\xi(f)$ is a centred Gaussian random variable, and its covariance structure is given by

$$\mathbb{E} \xi(f) \xi(g) = \int_{\mathbb{R}^n} f(x) g(x) \, dx, \quad f, g \in \mathcal{S}(\mathbb{R}^n).$$

In other words, the covariance is the $L^2(\mathbb{R}^n)$ inner product of the test functions.

Heuristically, white noise can be viewed as a universal source of randomness that is completely uncorrelated in space and in time. Its significance lies in the fact that it provides a natural way to model random perturbations in a continuum setting. The term “white” reflects the fact that the Fourier transform of its covariance is constant, so the noise has a flat power spectrum: all frequencies are present with equal intensity. Stochastic PDEs driven by white noise arise naturally as macroscopic or mesoscopic scaling limits of microscopic random systems. In this context, white noise appears because of its scale-invariant character: under suitable rescaling of space and time, its law remains unchanged. This invariance makes it the fundamental building block for modelling random fluctuations in large-scale limits.

Throughout, we denote spacetime points by $(t, x) \in \mathbb{R} \times \mathbb{R}^d$. A fundamental example of

an SPDE is the stochastic heat equation

$$(\partial_t - \Delta)\Phi = \xi, \quad \text{on } \mathbb{R}_+ \times \mathbb{R}^d, \quad (2.1)$$

where ξ is spacetime white noise on $\mathbb{R} \times \mathbb{R}^d$. This equation can be seen as a natural model for the diffusion of a field under random forcing. Since the driving noise is highly irregular, solutions typically cannot be interpreted pointwise and must be understood in a distributional sense.

Remark 2.1. Let us illustrate how the one-dimensional stochastic heat equation arises naturally from a simple physical model of a polymer immersed in a liquid. This example provides a clear intuition for why space-time white noise appears and shows how large-scale, macroscopic behaviour of random systems can emerge from microscopic randomness through appropriate scaling limits. Polymers, such as DNA or proteins, can be modelled as chains of beads connected by elastic springs, immersed in a liquid. Each bead is subject to two forces: the elastic pull from its neighbours and random collisions with the molecules of the surrounding fluid. We can formalize this picture using a discrete, microscopic model for the polymer. Denoting the position of bead i at time t by $\Psi(t, i)$, the dynamics are governed by the equation

$$\partial_t \Psi(t, i) = (\Psi(t, i+1) - \Psi(t, i)) + (\Psi(t, i-1) - \Psi(t, i)) + \zeta(t, i),$$

where the first and the second brackets on the right-hand side represent the approximately harmonic (i.e. proportional to the distance) forces exerted on i -th bead by the $i+1$ and $i-1$ beads, respectively, and $\zeta(t, i)$ represents the random kicks from the liquid. We assume that the random forces exerted by the environment acting on different beads are independent and identically distributed. We also assume that ζ exhibits only short range correlations in time. In a more compact form, the above equation reads

$$(\partial_t - \Delta_1)\Psi = \zeta \quad \text{on } \mathbb{R}_+ \times \mathbb{Z},$$

where Δ_1 is the nearest neighbour discrete Laplacian on $\mathbb{Z}$. To understand the large-scale, macroscopic behaviour of the polymer, we zoom out using a scaling transformation. Introduce a small parameter κ and define rescaled variables:

$$\Phi_\kappa(t, x) := \kappa^{1/2} \Psi(t/\kappa^2, x/\kappa), \quad \xi_\kappa(t, x) := \kappa^{-3/2} \zeta(t/\kappa^2, x/\kappa).$$

In these rescaled coordinates, the dynamics become

$$(\partial_t - \Delta_\kappa)\Phi_\kappa = \xi_\kappa \quad \text{on } \mathbb{R}_+ \times \kappa\mathbb{Z},$$

where Δ_κ is the nearest neighbour Laplacian on $\kappa\mathbb{Z}$. In the continuum limit $\kappa \searrow 0$ the discrete Laplacian Δ_κ converges to the continuum Laplacian Δ . Under our assumptions on the microscopic noise ζ , it can be shown that the rescaled forcing ξ_κ converges in law to space-time white noise ξ as $\kappa \searrow 0$. Consequently, the rescaled solution Φ_κ converges to a limiting process Φ_0 that satisfies the one-dimensional stochastic heat equation.

The nonlinear SPDEs we aim to study can be regarded as perturbations of the stochastic heat equation. To understand the singular nature of these equations, it is essential first to analyse the regularity properties of solutions to the stochastic heat equation. To quantify this regularity, we employ parabolic Hölder–Besov spaces $\mathcal{C}^\alpha = \mathcal{C}^\alpha(\mathbb{R} \times \mathbb{R}^d)$ defined with respect to the parabolic scaling. This scaling reflects the structure of the heat operator: one time derivative has the same order as two spatial derivatives. Accordingly, we measure distances using the parabolic metric

$$|(t, x), (s, y)| = |t - s|^{1/2} + |x - y|, \quad (t, x), (s, y) \in \mathbb{R} \times \mathbb{R}^d.$$

For $\alpha \in (0, 1)$, the space $\mathcal{C}^\alpha(\mathbb{R} \times \mathbb{R}^d)$ consists of continuous functions f such that

$$\|f\|_{\mathcal{C}^\alpha} := \sup_{(t, x)} |f(t, x)| + \sup_{(t, x) \neq (s, y)} \frac{|f(t, x) - f(s, y)|}{|(t, x) - (s, y)|^\alpha} < \infty.$$

Thus u is α -Hölder continuous with respect to the parabolic distance. Concretely, this means that u is α -Hölder in space and $\alpha/2$ -Hölder in time, reflecting the diffusive scaling of the heat equation. For $\alpha \in (1, 2)$ the space $\mathcal{C}^\alpha(\mathbb{R} \times \mathbb{R}^d)$ consists of bounded and continuous functions f continuously differentiable in space and such that

$$\|f\|_{\mathcal{C}^\alpha} := \sup_{(t, x)} |f(t, x)| + \sup_{(t, x) \neq (s, y)} \frac{|f(t, x) - f(s, y) - (x - y) \cdot (\nabla f)(s, y)|}{|(t, x) - (s, y)|^\alpha} + \|\nabla f\|_{\mathcal{C}^{\alpha-1}} < \infty.$$

For $\alpha \in (2, 3)$, the definition requires one time derivative and two spatial derivative. Hopefully, the reader will be able to generalize the definition to arbitrary positive non-integer α . Note that the time derivative ∂_t maps $\mathcal{C}^\alpha$ to $\mathcal{C}^{\alpha-2}$ and the spatial derivative ∇ maps $\mathcal{C}^\alpha$ to $\mathcal{C}^{\alpha-1}$. For negative non-integer α , the Hölder–Besov spaces can be defined recursively by

$$\mathcal{C}^\alpha(\mathbb{R} \times \mathbb{R}^d) = (\partial_t + 1 - \Delta) \mathcal{C}^{\alpha+2}(\mathbb{R} \times \mathbb{R}^d).$$

We say that $\phi \in \mathcal{S}'(\mathbb{R} \times \mathbb{R}^d)$ has regularity $\alpha \in \mathbb{R}$ if $\phi \in \mathcal{C}^\alpha(\mathbb{R} \times \mathbb{R}^d)$. This characterisation makes clear that applying the heat operator lowers regularity by two orders.

Let us list several standard facts about Besov spaces:

- (1) If $\alpha + \beta > 0$, the pointwise product extends continuously to a bilinear map

$$\mathcal{C}^\alpha \times \mathcal{C}^\beta \longrightarrow \mathcal{C}^{\alpha \wedge \beta}.$$

This condition $\alpha + \beta > 0$ is sharp as multiplying very irregular distributions is in general ill-defined.

- (2) Convolution with the heat kernel G , that is the Green function of the heat operator $\partial_t - \Delta$, gains two degrees of regularity in the parabolic Hölder scale. More precisely, the map

$$f \mapsto G * f : \mathcal{C}^\alpha \longrightarrow \mathcal{C}^{\alpha+2}$$

is continuous.

(3) If $\alpha \geq \beta$, there is a continuous embedding

$$\mathcal{C}^\alpha \hookrightarrow \mathcal{C}^\beta.$$

See, Sec. 7 and [BCD11, Sections 2.7–2.8], for the precise definition of the spaces $\mathcal{C}^\alpha$ and the proofs of the above facts. Note that we introduced the spaces $\mathcal{C}^\alpha$ only for non-integer α . For positive integer α the spaces $\mathcal{C}^\alpha$ are usually defined so that they coincide with the so-called Zygmund spaces (for example, $\mathcal{C}^1$ contains all Lipschitz functions).

Remark 2.2. Let θ denote the characteristic function of $[0, \infty)$. The function

$$\vartheta : (t, x^1, \dots, x^d) \mapsto \theta(t)\theta(x^1) \cdots \theta(x^d)$$

is bounded. Hence, we expect that it belongs to $\mathcal{C}^\alpha(\mathbb{R} \times \mathbb{R}^d)$ for every $\alpha < 0$. Since the Dirac delta on the spacetime $\mathbb{R} \times \mathbb{R}^d$ satisfies

$$\delta = \partial_t \partial_{x^1} \cdots \partial_{x^d} \vartheta$$

we conclude that δ belongs to $\mathcal{C}^\alpha(\mathbb{R} \times \mathbb{R}^d)$ for every $\alpha < -2 - d$. In other words, the regularity of the Dirac delta is slightly worse than minus the “effective” parabolic dimension of spacetime, which coincides with $2 + d$.

Since the covariance of the spacetime white noise coincides with the Dirac delta we expect that sample paths of the noise should have regularity α for all $\alpha < -(2+d)/2$. This heuristic is indeed correct locally: for every test function $\chi \in C_c^\infty(\mathbb{R} \times \mathbb{R}^d)$, we have $\chi\xi \in \mathcal{C}^\alpha(\mathbb{R} \times \mathbb{R}^d)$ almost surely for all $\alpha < -(2+d)/2$ (see e.g. [MWX16, Thm. 5]). Thus, the regularity of spacetime white noise is slightly worse than minus half of the “effective” parabolic dimension of spacetime.

Remark 2.3. Throughout, we write $f \in \mathcal{C}^{\beta-}$ to indicate that $f \in \mathcal{S}'$ is locally almost of regularity $\beta \in \mathbb{R}$, meaning that $\chi f \in \mathcal{C}^\alpha$ for every $\alpha < \beta$ and every $\chi \in C_c^\infty$.

In particular, since $\xi \in \mathcal{C}^{-1-d/2-}$ almost surely, the smoothing properties of the inverse of $\partial_t - \Delta$ suggest that a solution to the stochastic heat equation (2.1) satisfies $\Phi \in \mathcal{C}^{1-d/2-}$ almost surely. Consequently, in spatial dimension $d = 1$ the solution is Hölder continuous, whereas for $d > 1$ it remains a genuine distribution.

Exercise 2.1. Show that the law of the white noise ξ on $\mathbb{R} \times \mathbb{R}^d$ is invariant under the (natural generalization to distributions of the) scaling transformation $\kappa^{-(2+d)/2} \xi(t/\kappa^2, x/\kappa)$, where $\kappa > 0$.

3 Examples of singular stochastic PDEs

We now present several canonical examples of singular SPDEs. These equations arise in a variety of contexts, most prominently in nonequilibrium statistical physics, quantum field

theory, and materials science. Despite their different origins, they share a common structural feature: they are driven by spacetime white noise—an extremely irregular forcing term—and are coupled with nonlinear interactions. While the linear part alone already produces solutions of very low regularity, the presence of nonlinear terms leads to products of distributions that are not classically well defined. As a result, the equations are ill posed in the traditional analytical sense and require a suitable renormalization procedure to give them mathematical meaning.

KPZ Equation

One of the most well-known singular SPDE is the *Kardar–Parisi–Zhang (KPZ) equation* given by

$$(\partial_t - \Delta)\Phi = \xi + \lambda(\nabla\Phi)^2 - \infty \quad \text{on } \mathbb{R}_+ \times \mathbb{R}^d, \quad d \in \{1, 2\}, \quad (3.1)$$

where ξ denotes spacetime white noise, and the subtraction of ∞ corresponds to the necessary renormalization of the nonlinear term.

This equation models the stochastic growth of an interface: $\Phi(t, x)$ represents the height of a growing surface at time t and position x . Typical examples include deposition and erosion processes, flame fronts, or crystal growth—situations in which a surface evolves under the combined influence of deterministic smoothing and random forcing. The Laplacian term $\Delta\Phi$ encodes surface tension that smooths the interface, while the nonlinear term $(\nabla\Phi)^2$ captures the lateral growth, which tends to amplify small fluctuations in the slope. The noise term ξ represents random microscopic fluctuations in the growth process, such as random deposition or irregularities in the medium. From physical point of view the cases of one and two spatial dimensions are the most interesting.

Mathematically, the solution to the KPZ equation is expected to have the regularity not better than the regularity of the solution of the stochastic heat equation (the KPZ equation coincides with the stochastic heat equation for $\lambda = 0$). Thus, we expect $\Phi \in \mathcal{C}^{1-d/2-}$ almost surely. But then $\nabla\Phi \in \mathcal{C}^{-d/2-}$ has negative regularity and cannot be squared. This means that the nonlinear term $(\nabla\Phi)^2$ is ill-defined in a classical sense. The renormalization ∞ is required to subtract an infinite contribution arising from the square of the gradient.

Remark 3.1. The KPZ equation arises naturally as a *universal macroscopic scaling limit* of a wide class of interface growth models. Consider a generic growth model of the form

$$(\partial_t - \Delta)\Psi = \zeta + vN(\Psi), \quad \text{on } \mathbb{R}_+ \times \mathbb{R}^d, \quad (3.2)$$

where ζ is a smooth random noise with short-range correlations, and N is a nonlinear interaction term. We only assume basic symmetries for N , such as invariance under height shifts and spatial reflection.

In many physical systems, such as crystal growth or flame propagation, particles attach to the interface in a direction perpendicular to the local surface. As a consequence, the interface

advances in the normal direction, a phenomenon known as the *lateral growth mechanism*. If the interface is described as the graph of a height function $(x, \Psi(t, x))$, its unit normal vector is given by

$$n = \frac{(-\nabla\Psi, 1)}{\sqrt{1 + |\nabla\Psi|^2}}.$$

When the interface moves in the normal direction with speed v , the vertical component of this motion determines the evolution of the height function and yields

$$\partial_t\Psi = v\sqrt{1 + |\nabla\Psi|^2}.$$

This geometric relation expresses how the slope of the interface influences the effective vertical growth rate. Thus, a natural choice for N in (3.2) is

$$N(\Psi) = \sqrt{1 + |\nabla\Psi|^2}. \quad (3.3)$$

We then ask: what is the behaviour of the system under a *macroscopic rescaling* of space and time, possibly accompanied by a tuning of the nonlinearity? Suppose that $d = 1$. Under the scaling

$$\Phi_\kappa(t, x) = \kappa^{1/2}\Psi(t/\kappa^2, x/\kappa), \quad \xi_\kappa(t, x) = \kappa^{-3/2}\zeta(t/\kappa, x/\kappa),$$

with a weak nonlinearity $v = \tilde{v}\kappa^{1/2}$, one finds that, in the limit $\kappa \searrow 0$, the rescaled field with the average growth velocity removed, $\Phi_\kappa(t, x) - v_\kappa t$, converges to a solution of the KPZ equation. For a rigorous proof of this result, see [HS17, HQ18].

This is an example of a *macroscopic or mesoscopic scaling limit*: the microscopic details of the model largely disappear, and what remains is a universal effective equation. Remarkably, the limiting KPZ equation is driven by spacetime white noise and is the same across a broad class of microscopic models. This illustrates that singular SPDEs are not exotic or artificial—they are canonical objects describing the large scale behaviour of complex random systems. For an introduction to the concept of scaling limits in the weakly nonlinear regime in the context of singular SPDEs, see [Hai18].

Exercise 3.1. *Ignoring the singular nature of the KPZ equation, determine the average velocity v_ε such that, under the rescaling introduced above, the SPDE (3.2) with N given by (3.3) and $v = \lambda\kappa^{1/2}$ formally converges to (3.1). Note that, due to the need for renormalisation, the actual values of v and v_ε differ.*

Parabolic Stochastic Quantization Equations

Many important singular SPDEs arise naturally in the context of quantum field theory, where they serve as tools for rigorously constructing measures associated with quantum field models. To motivate this, let us first recall a classical fact from stochastic analysis regarding Langevin dynamics in finite dimensions.

Consider the stochastic differential equation

$$d\Phi_t(\phi) = dW_t - \nabla V(\Phi_t(\phi)) dt, \quad \Phi_0 = \phi,$$

where W_t is an n -dimensional Wiener process and $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is a sufficiently regular potential. Under mild assumptions on V , it is well known that the dynamics Φ_t preserves the measure

$$\mu(d\phi) = \frac{1}{\mathcal{Z}} \exp(-2V(\phi)) d\phi,$$

where $d\phi$ is the Lebesgue measure on $\mathbb{R}^n$ and $\mathcal{Z} > 0$ is a normalization constant making μ a probability measure. More precisely, let $(\mathcal{P}_t)_{t \geq 0}$ denote the Markov semigroup associated with the dynamics, defined by

$$(\mathcal{P}_t F)(\phi) = \mathbb{E} F(\Phi_t(\phi))$$

for bounded measurable functions $F : \mathbb{R}^n \rightarrow \mathbb{R}$. Then for all $t \geq 0$, we have

$$\mu(\mathcal{P}_t F) = \mu(F),$$

i.e., μ is invariant under the dynamics.

In infinite-dimensional settings, it is often more tractable to first construct the stochastic dynamics Φ_t and then use it to define an invariant measure. This approach is known as *stochastic quantization* and provides a probabilistic method for constructing measures corresponding to quantum field theories.

Dynamical Φ_d^4 Model. The dynamical Φ_d^4 model, also known as the parabolic stochastic quantization equation of the Φ_d^4 theory, is the parabolic SPDE

$$(\partial_t + 1 - \Delta)\Phi = \xi - \lambda\Phi^3 + \infty\Phi, \quad \text{on } \mathbb{R}_+ \times \mathbb{R}^d, \quad d \in \{2, 3\}, \quad (3.4)$$

where ξ denotes spacetime white noise. The term $\infty\Phi$ represents a (diverging) mass counterterm arising from renormalization. The dynamical Φ_d^4 model plays a fundamental role in constructive quantum field theory. It also appears in statistical mechanics, where it describes the mesoscopic behaviour of a ferromagnet near a critical phase transition.

Formally, the invariant measure associated with (3.4) is the Φ_d^4 Gibbs measure

$$\mu(d\phi) = \frac{1}{\mathcal{Z}} \exp \left(- \int_{\mathbb{R}^d} \left(\phi(1 - \Delta)\phi + \frac{\lambda}{2}\phi^4 - \infty\phi^2 \right) dx \right) d\phi, \quad (3.5)$$

where $d\phi$ denotes a (non-existent) Lebesgue measure on the space of distributions $\mathcal{S}'(\mathbb{R}^d)$.

To see the analogy with finite-dimensional Langevin dynamics, define formally

$$V(\phi) := \int_{\mathbb{R}^d} \left(\frac{1}{2}\phi(1 - \Delta)\phi + \frac{\lambda}{4}\phi^4 - \infty\phi^2 \right) dx.$$

Then its formal L^2 -gradient is

$$\nabla V(\phi) = (1 - \Delta)\phi + \lambda\phi^3 - \infty\phi,$$

so that (3.4) can be interpreted as an infinite-dimensional Langevin equation driven by an L^2 -cylindrical Wiener process.

From the linear theory one expects that solutions to the Φ_d^4 model satisfy $\Phi \in \mathcal{C}^{1-d/2-}$ almost surely. In dimensions $d \geq 2$ this regularity is negative. Consequently, the cubic term Φ^3 is not classically well defined and requires a suitable renormalization procedure, formally represented above by the infinite mass counterterm.

Dynamical Fractional $\Phi_{d,\sigma}^4$ Model. For $d \in \{2, 3\}$ and $\sigma > 0$, the fractional $\Phi_{d,\sigma}^4$ model is a non-local parabolic SPDE of the form

$$(\partial_t + 1 + (-\Delta)^{\sigma/2})\Phi = \xi - \lambda\Phi^3 + \infty\Phi \quad \text{on} \quad \mathbb{R}_+ \times \mathbb{R}^d, \quad (3.6)$$

where $(-\Delta)^{\sigma/2}$ denotes the fractional Laplacian of order σ , defined as a Fourier multiplier. Formally, the invariant measure of this equation coincides with (3.5) with $-\Delta$ replaced by $(-\Delta)^{\sigma/2}$. In particular, for $\sigma = 2$ one recovers the standard dynamical Φ_d^4 model.

To analyse the regularity of solutions, it is natural to work in fractional parabolic Hölder-Besov spaces $\mathcal{C}^\alpha$, defined using the fractional parabolic distance

$$|(t, x), (s, y)| = |t - s|^{1/\sigma} + |x - y|, \quad (t, x), (s, y) \in \mathbb{R} \times \mathbb{R}^d.$$

With respect to this scaling, spacetime white noise has regularity slightly worse than minus half of the “effective” dimension of spacetime, which is $\sigma + d$. In other words, $\xi \in \mathcal{C}^{-(\sigma+d)/2-}$ almost surely. The inverse of the operator $\partial_t + 1 + (-\Delta)^{\sigma/2}$ then improves regularity by σ , leading to $\Phi \in \mathcal{C}^{(\sigma-d)/2-}$ for solutions of the fractional $\Phi_{d,\sigma}^4$ model. In particular, the regularity is negative whenever $\sigma \leq d$, in which case the equation is singular.

Geometric Stochastic Heat Equation. Fix $n \in \mathbb{N}_+$. Let M be a submanifold of $\mathbb{R}^n$ and $\xi = (\xi^\gamma)_{\gamma \in \{1, \dots, n\}}$ be a collection of independent spacetime white noises and consider the SPDE

$$(\partial_t - \Delta)\Phi^\alpha = \sum_{\beta, \gamma=1}^n \Gamma_{\beta\gamma}^\alpha(\Phi) (\nabla\Phi^\beta) \cdot (\nabla\Phi^\gamma) + \sum_{\gamma=1}^n \sigma_\gamma^\alpha(\Phi) \xi^\gamma, \quad \text{on} \quad \mathbb{R}_+ \times \mathbb{R}^d,$$

where $\Phi = (\Phi^\alpha)_{\alpha \in \{1, 2, \dots, n\}}$ is a map from $\mathbb{R}^d$ to $M \subset \mathbb{R}^n$, $\Gamma_{\beta\gamma}^\alpha$ are the Christoffel symbols for the Levi-Civita connection on M , and $\{\sigma_\gamma\}_{\gamma=1}^n$ is a collection of vector fields tangent to M such that

$$\sum_{\gamma=1}^n \sum_{\beta=1}^n \sigma_\gamma^\alpha \sigma_\gamma^\beta X^\beta = X^\alpha$$

for all vectors X tangent to M . The geometric stochastic heat equation is a natural generalization of the classical stochastic heat equation. It is obtained by projecting the terms $\Delta\Phi + \xi$ from the standard equation onto the tangent space of M , ensuring that the evolving field remains on the submanifold at all times. The invariant measure of this equation corresponds to the so-called QFT sigma model. In dimension $d = 1$, this measure coincides with the law of a Brownian loop on M . See [BGHZ22] for more details.

Exercise 3.2. Suppose that σ_γ^α is smooth. Is the product $\sigma_\gamma^\alpha(\Phi) \xi^\gamma$ classically well-defined in dimension $d = 1$?

Exercise 3.3. Assuming the noise ξ is smooth, derive the form of the geometric heat equation in the case of a sphere, $M = \mathbb{S}^{n-1} \subset \mathbb{R}^n$.

Elliptic Stochastic Quantization

The elliptic stochastic quantization equation for the Φ_d^4 model is an elliptic SPDE of the form

$$(1 - \Delta)\Phi = \xi - \lambda\Phi^3 + \infty\Phi, \quad \text{on } \mathbb{R}^2 \times \mathbb{R}^d, \quad d \in \{2, 3\},$$

where ξ is the white noise on $\mathbb{R}^2 \times \mathbb{R}^d$. One shows that, formally, the solution restricted to a hypersurface of codimension two, $\Phi(0, 0, \cdot)$, is distributed according to the Φ_d^4 measure (3.5). See [ADV20] for more details about the elliptic stochastic quantization.

Exercise 3.4. What is the expected regularity of the solution? In which dimensions the equation is expected to be singular?

Exercise 3.5. Let Φ be the solution of $(1 - \Delta)\Phi = \xi$ on $\mathbb{R}^2 \times \mathbb{R}^d$. Show that $\Phi(0, 0, \cdot)$ is a Gaussian field over $\mathbb{R}^d$ with mean zero and variance $\frac{1}{4\pi}(1 - \Delta)^{-1}$.

4 Regularisation and renormalisation

All of the SPDEs introduced in the previous section are singular as soon as $\lambda \neq 0$. In the linear case $\lambda = 0$, the equations can be solved explicitly. However, the resulting solution already exhibits very low regularity. It is natural to seek solutions to the nonlinear equations in the same Hölder–Besov space in which the linear solution lives. As we have seen, this immediately leads to a difficulty: the nonlinear terms appearing in the equations are not classically well defined at that level of regularity.

The standard way to address this issue is to introduce an ultraviolet (UV) regularization—namely, to smooth the noise or the Green’s function at small spatial scales, or to discretize the equation. One then studies the regularized problem and attempts to pass to the limit as the regularization is removed. In general, the limit of the nonlinear terms appearing in the equation do not exist without further modification. The key idea is to perform

suitable renormalization of the nonlinear terms: one subtracts carefully chosen (typically divergent) constants that capture the small-scale singular behaviour. These subtractions compensate for the divergences and make it possible to obtain a well-defined, nontrivial limiting object. Without such renormalization, the limiting dynamic is either ill-defined or becomes trivial. For instance, in the dynamical Φ_2^4 model, as the mollifier is removed the solutions converge weakly to zero, since the cubic nonlinearity produces an increasingly strong damping effect [?].

Regularisation procedures

Discretisation. A natural way to regularize singular SPDEs such as the dynamical Φ_d^4 model (3.4) is to rewrite them on a finite lattice, say a toroidal lattice $\mathbb{T}_{\ell,\kappa}^d$ with spacings $\kappa \in (0, 1]$ and the total length $\ell \geq 1$. In this way we arrive at a finite system of SDEs of the form

$$d\Phi(t) = \kappa^{-d/2} d\hat{W}_\kappa(t) - (1 - \Delta_\kappa)\Phi(t)dt - \lambda\Phi(t)^3 dt + C_\kappa \Phi(t)dt, \quad (4.1)$$

where $\hat{W}_{\ell,\kappa}(t) = (\hat{W}_{\ell,\kappa}(t, x))_{x \in \mathbb{T}_{\ell,\kappa}^d}$ is a family of independent Wiener processes indexed by lattice sites,

$$(\Delta_\kappa f)(x) := \kappa^{-2} \sum_{\substack{y \in \mathbb{T}_{\ell,\kappa}^d \\ |y-x|=\kappa}} (f(y) - f(x))$$

denotes the nearest-neighbour Laplacian on $\mathbb{T}_{\ell,\kappa}^d$ and $C_\kappa \in \mathbb{R}$ is a counterterm that may depend on the lattice spacing $\kappa > 0$. By standard results about finite dimensional SDEs there exists a unique global periodic in space solution

$$\hat{\Phi}_{\ell,\kappa} : \mathbb{R}_+ \times \mathbb{T}_{\ell,\kappa}^d \rightarrow \mathbb{R}$$

to this equation with arbitrary initial condition ϕ . Moreover, the equation admits a unique explicit invariant measure

$$\mu_{\ell,\kappa}(d\phi) = \frac{1}{\mathcal{Z}_{\ell,\kappa}} \exp \left(- \frac{1}{\kappa^d} \sum_{x \in \mathbb{T}_{\ell,\kappa}^d} \left(\phi(1 - \Delta_\kappa)\phi + \frac{\lambda}{2} \phi^4 - C_\kappa \phi^2 \right) \right) \prod_{x \in \mathbb{T}_{\ell,\kappa}^d} d\phi(x), \quad (4.2)$$

which is absolutely continuous with respect to the finite product of the Lebesgue measures $d\phi(x)$ over lattice sites $x \in \mathbb{T}_{\ell,\kappa}^d$.

Let us denote by $\Phi_{\ell,\kappa} : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}$ any reasonable extension of $\hat{\Phi}_{\ell,\kappa}$ to continuum (say piecewise constant extension over blocks of length κ centred at lattice sites). The central question is whether one can choose the constants $(C_\kappa)_{\kappa \in (0,1]}$ in such a way that the limit

$$\Phi_\ell = \lim_{\kappa \searrow 0} \Phi_{\ell,\kappa}$$

exists as a tempered distribution and is not identically zero. If such a limit exists and is independent of the particular choice of the extension to the continuum involved in the definition

of $\Phi_{\ell,\kappa}$, then it is taken to define the solution of the corresponding singular SPDE—here, the dynamical Φ_d^4 model on $\mathbb{R}_+ \times \mathbb{T}_\ell^d$, where $\mathbb{T}_\ell^d$ is the d -dimensional torus of length ℓ . Next, interesting question one could study is the existence of the infinite volume limit

$$\Phi = \lim_{\ell \rightarrow \infty} \Phi_\ell.$$

Remark 4.1. Note that solving (4.1) directly in infinite volume—say on the full lattice $\mathbb{Z}^d$ with unit spacing—is far from straightforward. The main difficulty stems from the driving noise. If we denote by $\hat{W}(t) = (\hat{W}(t, x))_{x \in \mathbb{Z}^d}$ a family of independent Wiener processes, then

$$\sup_{x \in \mathbb{Z}^d} |\hat{W}(t, x)| \quad (4.3)$$

is almost surely infinite for every fixed $t > 0$. In other words, the noise is unbounded in space, and therefore the natural supremum norm is not suitable in infinite volume. To overcome this issue, one is forced to work in weighted spaces. More precisely, one considers norms of the form

$$\sup_{x \in \mathbb{Z}^d} w(x) |\hat{\Phi}(t, x)|,$$

where $w : \mathbb{Z}^d \rightarrow \mathbb{R}_+$ is a weight that decays polynomially as $|x| \rightarrow \infty$. Such weights accommodate the unavoidable spatial growth of a solution Φ induced by the unbounded noise. In particular, this setting prevents a direct application of the Banach fixed point theorem. Instead, one must derive suitable a priori estimates in weighted spaces and then rely on compactness and approximation arguments to construct solutions in infinite volume.

Exercise 4.1. Show that the measure (4.2) is invariant under the dynamic (4.1).

Exercise 4.2. Prove that (4.3) is almost surely infinite.

Mollification of noise. Let us describe another regularization procedure for parabolic SPDEs based on mollifying the driving noise. To this end pick a smooth, even, non-negative function $\vartheta \in C^\infty(\mathbb{R} \times \mathbb{R}^d)$ supported in the unit ball and normalized so that

$$\int \vartheta(t, x) dt dx = 1.$$

For $\kappa \in (0, 1]$, define the rescaled mollifier

$$\vartheta_\kappa(x) := [\kappa]^{-\sigma-d} \vartheta(t/\kappa^\sigma, x/\kappa).$$

With this scaling, ϑ_κ is supported in a parabolic ball of radius κ and converges to the Dirac distribution as $\kappa \searrow 0$. We then define the regularized noise by

$$\xi_{\ell,\kappa} := \vartheta_\kappa * \xi_\ell \in C^\infty(\mathbb{R} \times \mathbb{R}^d),$$

where $*$ denotes the spacetime convolution and ξ_ℓ is the periodisation in space of the space-time white noise ξ over $\mathbb{R} \times \mathbb{R}^d$ with period $\ell \geq 1$. For every $\kappa \in (0, 1]$, the field $\xi_{\ell, \kappa}$ is smooth, while as $\kappa \searrow 0$ we have $\xi_{\ell, \kappa} \rightarrow \xi_\ell$ in $\mathcal{S}'(\mathbb{R} \times \mathbb{R}^d)$ almost surely. We also have $\lim_{\ell \rightarrow \infty} \xi_\ell = \xi$. We now consider the family of regularized SPDEs

$$(\partial_t + 1 - \Delta)\Phi = \xi_{\ell, \kappa} - \lambda\Phi^3 + C_\kappa \Phi, \quad \text{on } \mathbb{R}_+ \times \mathbb{R}^d,$$

driven by the mollified noise $\xi_{\ell, \kappa}$ and involving a counterterm $C_\kappa \in \mathbb{R}$ that may depend on the regularization scale κ . For each $\ell \geq 1$ and $\kappa \in (0, 1]$, this is a classical (locally well-posed) equation with smooth forcing and we denote its solution by $\Phi_{\ell, \kappa}$. The central question is whether one can choose the constants $(C_\kappa)_{\kappa \in (0, 1]}$ in such a way that the limit

$$\lim_{\kappa \searrow 0} \Phi_{\ell, \kappa}$$

exists in a suitable topology and is nontrivial. If such a limit exists and is independent of the particular choice of mollifier ϑ , then it is taken to define the solution of the corresponding singular SPDE—here, the dynamical Φ_d^4 model on $\mathbb{R}_+ \times \mathbb{T}_\ell^d$.

Of course, the solution obtained by mollifying the noise should coincide with the solution obtained via a spatial discretisation. However, this agreement is only expected for appropriate choices of counterterms. Because there is some freedom in the choice of counterterms, both regularisation procedures lead to a one-parameter family of solutions.

Need for renormalisation

Before tackling nonlinear equations, it is instructive to understand where divergences actually come from. The phenomenon already appears at the level of the linear theory, once one attempts to form nonlinear expressions of the solution.

Let $\mathfrak{I}$ denote the stationary solution of the massive stochastic heat equation

$$(\partial_t + 1 - \Delta)\mathfrak{I} = \xi, \quad \text{on } \mathbb{R}_+ \times \mathbb{R}^d.$$

Formally, it can be written as

$$\mathfrak{I}(t, x) = \int G(t - s, x - y) \xi(ds, dy), \quad G(t - s, x - y) = \theta(t) e^{-(t-s)(1-\Delta)}(x, y),$$

that is, as the stochastic convolution of the noise with the heat semigroup.

A straightforward computation yields the covariance

$$\mathbb{E} \mathfrak{I}(t, x) \mathfrak{I}(t, y) = K(x - y) := \frac{1}{2} (1 - \Delta)^{-1}(x, y) \sim \begin{cases} 1, & d = 1, \\ \log(1/|x - y|), & d = 2, \\ |x - y|^{2-d}, & d > 2. \end{cases} \quad (4.4)$$

In particular, the covariance diverges at short distances when $d \geq 2$. This indicates that the pointwise square $\mathfrak{I}(t, x)^2$ should have infinite expectation in dimensions $d \geq 2$, and therefore cannot be meaningfully defined without modification.

To make this precise, consider a mollified version of the solution. Let $\mathfrak{I}_\kappa := M_\kappa * \mathfrak{I}$ denote the solution obtained by smoothing the noise at scale κ . Then

$$\mathbb{E} \mathfrak{I}_\kappa(t, x)^2 \sim \begin{cases} 1, & d = 1, \\ \log(1/\kappa), & d = 2, \\ \kappa^{2-d}, & d > 2 \end{cases}$$

Thus, as $\kappa \searrow 0$, the variance diverges logarithmically in dimension two and polynomially in higher dimensions. The divergence comes entirely from the small-scale behaviour of the noise.

This suggests subtracting the diverging constant and defining a renormalised square by

$$\mathbf{V} := \lim_{\kappa \searrow 0} \mathbf{V}_\kappa, \quad \mathbf{V}_\kappa(t, x) := \mathfrak{I}(t, x)^2 - c_\kappa, \quad c_\kappa := \mathbb{E} \mathfrak{I}_\kappa(t, x)^2.$$

This is the simplest example of Wick renormalisation. A direct computation shows that

$$\mathbb{E} \mathbf{V}(t, x) \mathbf{V}(t, y) = 2K(x - y)^2 \sim \begin{cases} 1, & d = 1, \\ \log(1/|x - y|)^2, & d = 2, \\ |x - y|^{4-2d}, & d > 2, \end{cases} \quad (4.5)$$

The function K^2 is locally integrable if and only if $d \leq 3$. Consequently, we expect that for $d \in 1, 2, 3$, the renormalized field $\mathbf{V}$ defines a random distribution whose action on test functions $f \in \mathcal{S}(\mathbb{R}^d)$ yields a square-integrable random variable with

$$\mathbb{E} \mathbf{V}(t, f)^2 = 2 \int f(x) K(x - y)^2 f(y) dx dy < \infty.$$

A more precise regularity estimate shows that the covariance K^2 belongs to $\mathcal{C}^\alpha$ for all $\alpha < 4 - 2d$, which suggests that almost surely $\mathbf{V} \in \mathcal{C}^{2-d-}$. In other words, the regularity of the renormalized square of $\mathfrak{I}$ is equal to twice the regularity of $\mathfrak{I}$ itself.

This illustrates a general principle: if two random distributions have negative regularities α_1 and α_2 , their product is typically ill-defined. However, after a suitable renormalization—usually by subtracting divergent constants—the product can often be meaningfully defined as a new distribution, whose regularity is given by the sum $\alpha_1 + \alpha_2$.

This simple computation captures the essence of renormalization: nonlinear expressions of rough random fields are divergent due to small-scale singularities, but carefully chosen subtractions of deterministic infinities yield well-defined, meaningful limiting objects.

Exercise 4.3. Prove (4.4) and (4.5).

Exercise 4.4. Define

$$\mathbb{V} := \lim_{\kappa \searrow 0} \mathbb{V}_\kappa, \quad \mathbb{V}_\kappa(t, x) := \mathbb{I}(t, x)^3 - 3c_\kappa \mathbb{I}(t, x), \quad c_\kappa := \mathbb{E} \mathbb{I}_\kappa(t, x)^2.$$

Show that

$$\mathbb{E} \mathbb{V}(t, x) \mathbb{V}(t, y) = 6K(x - y)^3.$$

In which dimensions $d > 0$ is $\mathbb{V}(t, \cdot)$ expected to be a well-defined distribution?

Subcriticality

A central concept in the theory of singular stochastic PDEs is that of subcriticality, which provides a criterion for whether a given equation can be meaningfully renormalized. Heuristically, for an SPDE with additive noise, the equation is called subcritical if the singularities created by the rough noise are not amplified by the nonlinearity. More precisely, after a suitable renormalization, the nonlinear terms should define distributions that are *strictly more regular* than the driving noise. In that case, one expects the solution of the full equation to be a perturbation of the solution to the corresponding linear stochastic heat equation.

Let us examine this principle in the case of the fractional $\Phi_{d,\sigma}^4$ model (3.6). From the linear theory, we expect $\Phi \in \mathcal{C}^{(\sigma-d)/2-}$. If $\sigma \leq d$, then the solution has negative regularity and is therefore a genuine distribution. Nevertheless, as suggested by the Wick renormalization argument from the previous section, one may hope to define a suitably renormalized cube Φ^3 with regularity $\mathcal{C}^{3(\sigma-d)/2-}$. The equation is subcritical if the cubic term is strictly more regular than the noise. Since spacetime white noise has regularity $-(\sigma + d)/2-$, this leads to the condition

$$3(\sigma - d)/2 > -(\sigma + d)/2.$$

This is equivalent to $\sigma > d/2$. Thus, for $\sigma \in (d/2, d]$, the model is both singular (because $\sigma \leq d$) and subcritical (because $\sigma > d/2$). In the classical case $\sigma = 2$, this yields:

- $d = 1$: the equation is not singular;
- $d = 2, 3$: the equation is singular but subcritical;
- $d = 4$: the equation is critical;
- $d > 4$: the equation is supercritical.

At present, a robust and systematic theory is available only in the subcritical regime. In the critical case, the non-linearities are expected to be roughly as singular as the driving noise, and new ideas would be required to construct solutions. In the supercritical case, we usually expect the solutions to be Gaussian.

The same scaling heuristic applies to other models. For example, the KPZ equation is subcritical in $d = 1$, critical in $d = 2$, and supercritical in higher dimensions. A more involved analysis shows that the geometric stochastic heat equation follows the KPZ pattern. In that case, although the noise is multiplicative, the solution can still be viewed—at small scales—as

a perturbation of the stochastic heat equation, with appropriately modified behaviour at larger scales.

Exercise 4.5. *Study the behaviour of the fractional $\Phi_{d,\sigma}^4$ model under the rescaling transformation*

$$\Phi_\kappa(t, x) = \kappa^{(d-\sigma)/2} \Phi(\kappa^2 t, \kappa x), \quad \xi_\kappa(t, x) = \kappa^{(d+\sigma)/2} \xi(\kappa^2 t, \kappa x).$$

Note that for small $\kappa > 0$ this transformation corresponds to zooming in. Show that, formally, the nonlinearity of the rescaled equation vanishes in the limit $\kappa \searrow 0$ in the subcritical case. Thus, in this regime the system is expected to be weakly interacting at small scales.

Challenges

The study of singular SPDEs presents several fundamental challenges, which are typically addressed in sequence:

- **Local solution theory.** Develop a well-posedness theory on short (possibly random) time intervals. Construct maximal solutions up to a possible blow-up.
- **Global solution theory.** Exploit dissipative structures—such as the cubic damping term in the dynamical Φ^4 model—to derive a priori bounds. Use these estimates to rule out blow-up and extend solutions globally in time. Subsequently, analyse infinite-volume limits and construct solutions on the full space.
- **Construction of quantum field theory (QFT) measures.** Study discretized versions of stochastic quantization equations and prove uniform a priori estimates. Use these bounds to establish tightness of the associated invariant measures. Passing to the limit yields a rigorous construction of the continuum QFT measure.
- **Analysis of invariant measures.** Prove structural properties such as invariance under Euclidean transformations, uniqueness of the limiting measure, and exponential decay of correlations.

Among the main open problems are the development of a global solution theory and the construction of QFT measures for models lacking strong damping mechanisms (for example, Yang–Mills theories), as well as the ultimate challenge of solving critical SPDEs.

In these notes, we present the main ideas underlying the local solution theory for subcritical equations.

5 Instructive toy model

We shall concentrate on a concrete example of a singular SPDE, namely the elliptic problem

$$(1 - \Delta)^{\sigma/2} \Phi = \xi + \lambda \Phi^3 - \infty \Phi, \quad \text{on } \mathbb{R}^d, \quad d = 5, \quad (5.1)$$

where ξ denotes the 2π -periodisation of white noise on $\mathbb{R}^d$. We fix $d = 5$ and assume $\sigma \in (d/3, d/2]$, which corresponds precisely to the full subcritical regime. Although (5.1) is not derived directly from a standard physical model, it serves as a particularly transparent and flexible testing ground for the analysis of singular SPDEs and their renormalization. By imposing periodicity on the noise, we place the problem in a finite-volume setting. This allows us to focus solely on the renormalization procedure itself, avoiding the further technical complications that arise in infinite-volume analysis. For $\sigma = 2$, the equation formally resembles the stochastic quantization equation associated with the Φ_3^4 model. However, due to the imposed periodicity of the noise in all spatial directions, the resulting solution is not related to the Φ_3^4 Euclidean quantum field measure.

Exercise 5.1. *Using the formal argument from the previous section convince yourself that the SPDE (5.1) is subcritical for $\sigma \in (d/3, d/2]$.*

We rewrite the singular SPDE (5.1) in the regularized mild form

$$\Phi = G * F_\kappa[\Phi], \quad \kappa \in (0, 1], \quad (5.2)$$

where $G \in L^1(\mathbb{R}^d)$ is the fundamental solution for the pseudo-differential operator $(1 - \Delta)^{\sigma/2}$. The functional $F_\kappa[\varphi]$, called the force, is defined by

$$F_\kappa[\varphi](x) := \xi_\kappa(x) + \lambda\varphi(x)^3 + \sum_{i=1}^{i_\sharp} \lambda^i c_\kappa^{(i)} \varphi(x), \quad (5.3)$$

where ξ_κ is the mollification of the periodic noise ξ , $i_\sharp := \lfloor \sigma/(3\sigma - d) \rfloor$ and the parameters $c_\kappa^{(i)} \in \mathbb{R}$ depending on the UV cutoff κ are called the counterterms. These constants are introduced to cancel the divergences arising from the nonlinear interaction in the limit $\kappa \searrow 0$. The number of such counterterms required to renormalize the cubic nonlinearity increases without bound as $\sigma \searrow d/3$. Consequently, the renormalization problem becomes progressively more intricate near the critical threshold $\sigma = d/3$, where the equation ceases to be subcritical.

We can now formulate the main result of these notes, which establishes that, after suitable renormalization, the family of regularized equations (5.2) admits a well-defined and nontrivial limit as $\kappa \searrow 0$.

Theorem 5.1. *Let $d = 5$ and $\sigma \in (d/3, d/2]$. There exist a choice of counterterms and random variables $\lambda_\star \in [0, 1]$ and $\Phi_0 \in \mathcal{S}'(\mathbb{R}^d)$ such that:*

- (0) *for all $\lambda \in [-\lambda_\star, \lambda_\star]$ and $\kappa \in (0, 1]$ there exists a solution $\Phi_\kappa \in C^\infty(\mathbb{R}^d)$ of (5.2),*
- (1) *$\Phi_0 = \lim_{\kappa \searrow 0} \Phi_\kappa$ almost surely in $\mathcal{S}'(\mathbb{R}^d)$,*
- (2) *$\mathbb{E}(\lambda_\star^{-n}) < \infty$ for every $n \in \mathbb{N}_+$.*

Note that λ , the prefactor of the non-linear term in the equation, is assumed to be random and sufficiently small. In the case of parabolic equations the method discussed in these notes can be used to construct a solution in a sufficiently small time interval without any assumption about the strength of the non-linearity. For concreteness suppose that $d = 3$, $\sigma \in (d/2, d]$. Denote by ξ the periodisation in space with the period 2π of the spacetime white noise on $\mathbb{R} \times \mathbb{R}^d$ and let ξ_κ denote its mollification at scale $\kappa \in (0, 1]$. Consider the following fractional parabolic SPDE

$$\partial_t + (-\Delta)^\sigma \Phi = \xi_\kappa + \lambda \Phi^3 + \sum_{i=1}^{i_\#} \lambda^i c_\kappa^{(i)} \Phi, \quad \text{on } \mathbb{R}_+ \times \mathbb{R}^d, \quad (5.4)$$

where $i_\# := \lfloor \sigma/(2\sigma - d) \rfloor$.

Theorem 5.2. *Let $d = 3$ and $\sigma \in (d/2, d]$. For $\kappa \in (0, 1]$ let $\Phi_\kappa \in C(\mathbb{R}_+, C^\infty(\mathbb{R}^d))$ be the unique mild solution of (5.4) with a suitable initial condition. There exists a choice of counterterms, a stopping time $T_\star \in \mathbb{R}_+$ and a random variable $\Phi \in \mathcal{S}'(\mathbb{R}_+ \times \mathbb{R}^d)$ such that:*

- (1) *for all $T > 0$, $\lim_{\kappa \searrow 0} \Phi_\kappa = \Phi$ almost surely in $\mathcal{S}'((0, T) \times \mathbb{R}^d)$ on the event $\{T_\star < T\}$,*
- (2) *$\mathbb{E}(T_\star^{-n}) < \infty$ for every $n \in \mathbb{N}_+$.*

6 Da Prato-Debussche regime

Let us now examine how the analytic difficulty of the problem depends on the parameter $\sigma > 0$. As σ decreases, the smoothing effect of the fractional Laplacian weakens, and new challenges emerge each time a critical regularity threshold is crossed. We begin by outlining the proof of the main result in the simplest setting, commonly referred to as the Da Prato–Debussche regime, following the classical approach of Da Prato and Debussche [DPD03]. This regime serves as a conceptual baseline for understanding the limitations of classical techniques and for motivating the more sophisticated methods required in the more singular case, such as the Φ_3^4 model.

Remark 6.1. In what follows, $\alpha = \sigma - d/2 - \varepsilon$ denotes the expected regularity of the solution of (5.2) and $\varepsilon \in (0, \infty)$ is assumed to be sufficiently small.

Using the above-mentioned facts one easily shows that for $\sigma \in (d/2, \infty)$ the equation we study, $\Phi = G * (\xi - \lambda \Phi^3)$, is not singular and can be solved in a Hölder space $\mathcal{C}^\alpha$ with $\alpha > 0$ using the Banach fixed-point theorem. In this regime no regularization is needed. Let us see what happens for smaller values of σ .

Proposition 6.2. *The statement of Theorem 5.1 holds true for $\sigma \in (5d/12, d/2]$.*

Sketch of the proof. Recall that we want to solve the equation

$$\Phi_\kappa = G * (\xi_\kappa + \lambda \Phi_\kappa^3 + \lambda c_\kappa^{(1)} \Phi_\kappa) \quad (6.1)$$

and prove the existence of the limit $\lim_{\kappa \searrow 0} \Phi_\kappa$. We have $G * \xi_0 \in \mathcal{C}^\alpha$ and we expect that $\Phi_0 \in \mathcal{C}^\alpha$. But since $\alpha < 0$ the cube Φ_0^3 is classically ill-defined. We shall employ the so-called Da Prato–Debussche trick [DPD03]. Introducing the notation $\bullet_\kappa := \xi_\kappa$ and $x \longmapsto y := G(x - y)$ we define $\mathfrak{l}_\kappa = G * \bullet_\kappa := G * \xi_\kappa$ and make the following ansatz for the solution of our equation

$$\Phi_\kappa = \mathfrak{l}_\kappa + \Psi_\kappa.$$

As we shall see, the remainder Ψ_0 has much better regularity than $\mathfrak{l}_0, \Phi_0 \in \mathcal{C}^\alpha$. Using (6.1) one shows that Ψ_κ satisfies the following equation

$$\Psi_\kappa = \lambda G * (\Psi_\kappa^3 + 3\Psi_\kappa^2 \mathfrak{l}_\kappa + 3\Psi_\kappa \mathfrak{V}_\kappa + \mathfrak{V}_\kappa^3), \quad (6.2)$$

where

$$\mathfrak{V}_\kappa := (\mathfrak{l}_\kappa)^2 + c_\kappa^{(1)}/3, \quad \mathfrak{V}_\kappa^3 := (\mathfrak{l}_\kappa)^3 + c_\kappa^{(1)} \mathfrak{l}_\kappa, \quad c_\kappa^{(1)} := -3 \mathbb{E} (\mathfrak{l}_\kappa)^2$$

for all $\kappa \in (0, 1]$.

The list of trees $(\mathfrak{l}_\kappa, \mathfrak{V}_\kappa, \mathfrak{V}_\kappa^3)$ is called the enhanced noise. Of course the products appearing in the definitions of the above trees are ill-defined deterministically for $\kappa = 0$. However, the point is that the trees are simple and quite explicit objects. It turns out that it is possible to control the convergence of the enhanced noise as $\kappa \searrow 0$ by studying its covariance. One shows that with the above choice of the counterterm the limit

$$\lim_{\kappa \searrow 0} (\mathfrak{l}_\kappa, \mathfrak{V}_\kappa, \mathfrak{V}_\kappa^3) =: (\mathfrak{l}_0, \mathfrak{V}_0, \mathfrak{V}_0^3) \in \mathcal{C}^\alpha \times \mathcal{C}^{2\alpha} \times \mathcal{C}^{3\alpha} \quad (6.3)$$

exists almost surely and there exists a random variable $R \in [1, \infty]$ such that $\mathbb{E} R^n < \infty$ for all $n \in \mathbb{N}_0$ and

$$\|\mathfrak{l}_\kappa\|_{\mathcal{C}^\alpha} \vee \|\mathfrak{V}_\kappa\|_{\mathcal{C}^{2\alpha}} \vee \|\mathfrak{V}_\kappa^3\|_{\mathcal{C}^{3\alpha}} \leq R \quad (6.4)$$

for all $\kappa \in [0, 1]$. We call the above bounds the stochastic estimates for the enhanced noise.

To proceed we rewrite (6.2) as the fixed point equation of the map

$$\mathbf{Q}[\psi] \equiv \mathbf{Q}[\psi; \mathfrak{l}, \mathfrak{V}, \mathfrak{V}^3] := \lambda G * (\psi^3 + 3\psi^2 \mathfrak{l} + 3\psi \mathfrak{V} + \mathfrak{V}^3).$$

We claim that the map

$$\mathbf{Q}: \mathcal{C}^{3\alpha+\sigma} \rightarrow \mathcal{C}^{3\alpha+\sigma}$$

is well defined for all $(\mathfrak{l}, \mathfrak{V}, \mathfrak{V}^3) \in \mathcal{C}^\alpha \times \mathcal{C}^{2\alpha} \times \mathcal{C}^{3\alpha}$. The above choice of the domain of $\mathbf{Q}$ is dictated by the regularity of $G * \mathfrak{V}^3 \in \mathcal{C}^{3\alpha+\sigma}$. Note that $3\alpha + \sigma > 0$. Hence, ψ in the domain of $\mathbf{Q}$ is a Hölder continuous function and the products

$$\psi^2, \psi^3 \in \mathcal{C}^{3\alpha+\sigma} \subset \mathcal{C}^{3\alpha}$$

are well defined. In order to make sure that the products

$$\psi^2 \mathfrak{l} \in \mathcal{C}^\alpha \subset \mathcal{C}^{3\alpha} \quad \psi \mathfrak{V} \in \mathcal{C}^{2\alpha} \subset \mathcal{C}^{3\alpha}$$

are well defined we have to check whether $(3\alpha + \sigma) + l\alpha > 0$ for $l \in \{1, 2\}$. Since $\alpha < 0$ it is enough to check this for $l = 2$. We obtain the condition $5\alpha + \sigma > 0$, which is equivalent to $\sigma > 5d/12$. Consequently, all of the products are well defined and belong to $\mathcal{C}^{3\alpha}$. As a result, the map $\mathbf{Q} : \mathcal{C}^{3\alpha+\sigma} \rightarrow \mathcal{C}^{3\alpha+\sigma}$ is well defined. $\square$

Exercise 6.1. *Complete the deterministic part of the above proof. Assume that the maps in Items (1), (2), (3) above have norms bounded by some constant $c \in [1, \infty)$. For $R \in [1, \infty)$ let $\mathcal{B}_R$ be the closed ball in $\mathcal{C}^{3\alpha+\sigma}$ of radius R and let $\lambda_* := 1/(100c^3R^2)$. For all $\kappa \in [0, 1]$ let $\mathbf{Q}_\kappa[\psi] := \mathbf{Q}[\psi; \mathbf{I}_\kappa, \mathbf{V}_\kappa, \mathbf{V}_\kappa]$. Assume that (6.4) holds true. Prove that for $\lambda \in [-\lambda_*, \lambda_*]$ and $\kappa \in [0, 1]$ the map $\mathbf{Q}_\kappa : \mathcal{B}_R \rightarrow \mathcal{B}_R$ is well defined and is a contraction with the Lipschitz constant less than $1/2$. Conclude that for all $\kappa \in [0, 1]$ the map $\mathbf{Q}_\kappa$ has a unique fixed point in $\mathcal{B}_R$ denoted by Ψ_κ . Next, using (6.3) show that for all $\psi \in \mathcal{B}_R$ it holds that $\lim_{\kappa \searrow 0} \mathbf{Q}_\kappa[\psi] = \mathbf{Q}_0[\psi]$. Conclude that $\lim_{\kappa \searrow 0} \Psi_\kappa = \Psi_0 \in \mathcal{C}^{3\alpha+\sigma}$ and $\lim_{\kappa \searrow 0} \Phi_\kappa = \Phi_0 \in \mathcal{C}^\alpha$.*

The method of the above proof does not work for $\sigma \leq 5d/12$ because the product of Ψ_0 and $\mathbf{V}_0$ is no longer well defined classically. The problem is that the Da Prato–Debussche remainder Ψ_0 does not have enough regularity. To proceed we apply the Da Prato–Debussche trick once again to remove the most singular tree $\mathbf{V}_\kappa$ on the right-hand side of (6.2). To this end, we make the following ansatz

$$\Phi_\kappa = \mathbf{I}_\kappa + \Psi_\kappa = \mathbf{I}_\kappa + \lambda \mathbf{V}_\kappa + \tilde{\Psi}_\kappa, \quad \mathbf{V}_\kappa = G * \mathbf{V}_\kappa.$$

The SPDE (6.1) implies that the remainder $\tilde{\Psi}_\kappa$ satisfies the equation

$$\tilde{\Psi}_\kappa = \lambda G * \left((\tilde{\Psi}_\kappa + \lambda \mathbf{V}_\kappa)^3 + 3(\tilde{\Psi}_\kappa + \lambda \mathbf{V}_\kappa)^2 \mathbf{I}_\kappa + 3\tilde{\Psi}_\kappa \mathbf{V}_\kappa + 3\lambda \mathbf{V}_\kappa \mathbf{V}_\kappa \right), \quad (6.5)$$

where the new tree is defined by

$$\mathbf{V}_\kappa := \mathbf{V}_\kappa \mathbf{V}_\kappa.$$

Suppose that $\sigma \in (2d/5, 5d/12]$. Even though the product in the definition of the above tree is ill-defined deterministically no renormalization is necessary. One shows that

$$\lim_{\kappa \searrow 0} \mathbf{V}_\kappa =: \mathbf{V}_0 \in \mathcal{C}^{2\alpha}$$

exists almost surely and there exists a random variable $R \in [1, \infty]$ such that $\mathbb{E}R^n < \infty$ for all $n \in \mathbb{N}_0$ and

$$\|\mathbf{V}_\kappa\|_{\mathcal{C}^{2\alpha}} \leq R$$

for all $\kappa \in [0, 1]$.

Proposition 6.3. *The statement of Theorem 5.1 holds true for $\sigma \in (2d/5, 5d/12]$.*

Exercise 6.2. *Reformulate (6.5) as a fixed-point problem in a ball in $\mathcal{C}^{2\alpha+\sigma}$ and prove the above proposition.*

It turns out at $\sigma = 2d/5$ the Da Prato–Debussche method breaks down completely. Since $\mathbf{V}_0 \in \mathcal{C}^{2\alpha}$ with $\alpha = \sigma - d/2 - \varepsilon$ the term $\tilde{\Psi}_0 \mathbf{V}_0$, which cannot be removed using Da Prato–Debussche trick, can at best be defined as an element of $\mathcal{C}^{2\alpha}$. But then one can only hope that $G * (\tilde{\Psi}_0 \mathbf{V}_0) \in \mathcal{C}^{2\alpha+\sigma}$. Thus, we should expect that $\tilde{\Psi}_0 \in \mathcal{C}^{2\alpha+\sigma}$. However, for $\tilde{\Psi}_0 \in \mathcal{C}^{2\alpha+\sigma}$ the product $\tilde{\Psi}_0 \mathbf{V}_0$ is well defined only if $(2\alpha + \sigma) + 2\alpha \geq 0$, which implies that $\sigma > 2d/5$.

7 Paracontrolled approach

For $\sigma = 2d/5$, controlling the product $\tilde{\Psi}_0 \mathbf{V}_0$ and closing the estimates requires additional structural information about $\tilde{\Psi}_0$ that cannot be captured solely in terms of its regularity. The main goal of this section is to establish the following proposition.

Proposition 7.1. *The statement of Theorem 5.1 holds true for $\sigma = 2d/5 = 2$.*

To this end, we employ the paracontrolled framework for singular SPDEs, originally introduced by Gubinelli, Imkeller and Perkowski in [GIP15]. The presentation given here adapts the paracontrolled ansatz proposed in [CC18], which was developed to study the dynamical Φ_3^4 model, to the present elliptic setting.

Before outlining the proof, we introduce a precise definition of Hölder–Besov spaces, making rigorous the informal discussion from Sec. 2. The key idea is to measure regularity by decomposing a distribution into its frequency components and controlling how these components behave at different scales.

Definition 7.2 (Littlewood–Paley blocks). Let $\chi \in C_c^\infty(\mathbb{R}^d)$ be a radial function supported in the annulus $\{\omega \in \mathbb{R}^d \mid \frac{3}{4} \leq |\omega| \leq \frac{8}{3}\}$, and let $\chi_{-1} \in C_c^\infty(\mathbb{R}^d)$ be a smooth low-frequency cutoff supported in $\{\omega \in \mathbb{R}^d \mid |\omega| \leq \frac{4}{3}\}$ such that

$$\sum_{i=-1}^{\infty} \chi_i = 1,$$

where $\chi_i(\omega) := \chi(2^{-i}\omega)$ for $i \in \mathbb{N}_0$. The *Littlewood–Paley blocks* $(\Delta_i)_{i \geq -1}$ are defined by

$$\Delta_i f := \mathcal{F}^{-1}(\chi_i \mathcal{F} f)$$

for $f \in \mathcal{S}'(\mathbb{R}^d)$, where $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the Fourier and inverse Fourier transforms, respectively.

The intuition is that $\Delta_j f$ isolates the portion of f whose Fourier modes live at frequencies $|\omega| \sim 2^j$. Thus each block captures the behavior of f at a specific spatial scale 2^{-j} . The low-frequency block $\Delta_{-1} f$ describes the large-scale component. Since the cutoffs form a partition of unity in Fourier space, one formally has

$$f = \sum_{j \geq -1} \Delta_j f$$

which decomposes f into contributions from all dyadic scales. Regularity can therefore be read off from how fast $\|\Delta_j f\|_{L^\infty(\mathbb{R}^d)}$ decays as $j \rightarrow \infty$.

Definition 7.3 (Hölder-Besov space). For $\alpha \in \mathbb{R}$, the *Hölder-Besov space* $\mathcal{C}^\alpha(\mathbb{R}^d)$ is defined as the completion of the space of test functions $f \in \mathcal{S}(\mathbb{R}^d)$ equipped with the norm

$$\|f\|_{\mathcal{C}^\alpha} := \sup_{i \geq -1} 2^{\alpha i} \|\Delta_i f\|_{L^\infty(\mathbb{R}^d)} < \infty,$$

where $(\Delta_i)_{i \geq -1}$ are the Littlewood-Paley blocks.

The factor $2^{\alpha i}$ compensates for the scaling of the i -th block. Roughly speaking, if $|\Delta_i f|_{L^\infty}$ decays like $2^{-\alpha i}$ as $i \rightarrow \infty$, then f behaves like a function with α derivatives. If $\alpha > 0$, this definition coincides with classical Hölder regularity. If $\alpha < 0$, the space allows distributions whose high-frequency components grow at a controlled rate.

Exercise 7.1. Prove that the d -dimensional Dirac delta belongs to $\mathcal{C}^{-d-\varepsilon}$ for all $\varepsilon > 0$.

Exercise 7.2. Let $\alpha \in (0, 1)$. Show that $\mathcal{C}^\alpha$ norm is equivalent to the standard α -Hölder norm. *Hint:* The Hölder norm of u can be controlled in terms of L^∞ -norm and $|u(x) - u(y)|$ for $|x - y| \leq 1$. Study separately the two cases : $2^j \leq |x - y|$ and $2^j > |x - y|$. Note that $\int \chi_j(x) dx = 0$ for all $j \geq 0$.

Exercise 7.3. Show that

$$\begin{aligned} \|u\|_\alpha &\lesssim \|u\|_\beta && \text{for } \alpha \leq \beta, \\ \|u\|_{L^\infty} &\lesssim \|u\|_\alpha && \text{for } \alpha > 0, \\ \|u\|_\alpha &\lesssim \|u\|_{L^\infty} && \text{for } \alpha \leq 0. \end{aligned}$$

The dyadic viewpoint is particularly powerful for nonlinear analysis, since products of distributions can be studied by analysing how their frequency components interact. To make this interaction precise, we decompose products into components corresponding to different frequency regimes.

Definition 7.4. For $u, v \in \mathcal{S}(\mathbb{R}^d)$, define the paraproduct

$$u \prec v = \sum_{\substack{i, j \geq -1 \\ i \leq j-2}} \Delta_i u \Delta_j v$$

and the resonant product

$$u \odot v = \sum_{\substack{i, j \geq -1 \\ |i-j| \leq 1}} \Delta_i u \Delta_j v.$$

We denote their sum by $u \preceq v = u \prec v + u \odot v$.

Remark 7.5. Note that we have

$$uv = u \prec v + u \odot v + v \succ v.$$

Remark 7.6. Observe that for any fixed $k \geq -1$ there are only finitely many terms contributing to the sum

$$\Delta_k \sum_{\substack{i,j \geq -1 \\ i \leq j-2}} \Delta_i u \Delta_j v.$$

Consequently, one can define the paraproduct of arbitrary Schwartz distributions. Since, in general, infinitely many terms contribute to the sum

$$\Delta_k \sum_{\substack{i,j \geq -1 \\ |i-j| \leq 1}} \Delta_i u \Delta_j v,$$

extending the definition of the resonant product is more delicate and requires suitable assumptions on the regularities of the distributions involved.

The paraproduct $u \prec v$ captures the interaction between low frequencies of u and high frequencies of v . In this regime, the low-frequency factor behaves essentially like a bounded coefficient, so the product is relatively well behaved. The resonant term $u \odot v$ describes interactions between comparable frequencies. This is the delicate part: when both factors are rough, their high-frequency components may combine to produce an ill-defined product. In fact, the condition $\alpha + \beta > 0$ in the estimates below reflects exactly the threshold for such interactions to remain meaningful.

Lemma 7.7 (Paraproduct Estimates). *For all $\alpha < 0$ and $\beta \in \mathbb{R}$, the paraproduct satisfies*

$$\begin{aligned} \|f \prec g\|_{\mathcal{C}^\beta} &\lesssim \|g\|_{L^\infty} \|g\|_{\mathcal{C}^\beta}, \\ \|f \prec g\|_{\mathcal{C}^{\alpha+\beta}} &\lesssim \|f\|_{\mathcal{C}^\alpha} \|g\|_{\mathcal{C}^\beta} \end{aligned}$$

uniformly over $f, g \in \mathcal{S}'(\mathbb{R}^d)$.

Furthermore, for all $\alpha, \beta \in \mathbb{R}$ such that $\alpha + \beta > 0$, then the resonant product satisfies

$$\|f \odot v\|_{\mathcal{C}^{\alpha+\beta}} \lesssim \|f\|_{\mathcal{C}^\alpha} \|g\|_{\mathcal{C}^\beta}$$

uniformly over $f, g \in \mathcal{S}(\mathbb{R}^d)$. In particular, the resonant product extends to a continuous map from $\mathcal{C}^\alpha \times \mathcal{C}^\beta$ to $\mathcal{C}^{\alpha+\beta}$.

Proof. See [BCD11, Sections 2.7–2.8]. □

Exercise 7.4. *Show that for all $\alpha, \beta \in \mathbb{R}$ such that $\alpha + \beta > 0$, $\|fg\|_{\mathcal{C}^{\alpha \wedge \beta}} \lesssim \|f\|_{\mathcal{C}^\alpha} \|g\|_{\mathcal{C}^\beta}$.*

Definition 7.8. For $f, g, h \in \mathcal{S}(\mathbb{R}^d)$, we define the commutator

$$C(f, g, h) = (f \prec g) \odot h - f(g \odot h).$$

Lemma 7.9. *The commutator C extends to a bounded trilinear map from $\mathcal{C}^\alpha \times \mathcal{C}^\beta \times \mathcal{C}^\gamma$ to $\mathcal{C}^{\alpha+\beta+\gamma}$, provided that $\alpha \in (0, 1)$ and $\beta + \gamma < 0$ while $\alpha + \beta + \gamma > 0$.*

Proof. See [GIP15, Lemma 2.4]. $\square$

Note that for $\beta + \gamma < 0$, the individual terms appearing in the definition of the commutator are not regular enough to be controlled separately. The crucial observation, however, is that while each term is too singular on its own, their difference is not. The commutator isolates a cancellation mechanism between frequency interactions, and it is this cancellation that restores regularity. This structural gain is what allows paracontrolled expansions to go beyond the Da Prato–Debussche threshold.

Let us now turn to the proof of Proposition 7.1. Our goal is to solve the mild formulation of the equation

$$(1 - \Delta)\Phi_\kappa = \xi_\kappa + \lambda \Phi_\kappa^3 + (\lambda c_\kappa^{(1)} + \lambda^2 c_\kappa^{(2)}) \Phi_\kappa \quad (7.1)$$

and to establish the existence of the limit $\lim_{\kappa \searrow 0} \Phi_\kappa$. We decompose the solution according to

$$\Phi_\kappa = \mathfrak{I}_\kappa + \Psi_\kappa = \mathfrak{I}_\kappa + \lambda \mathfrak{V}_\kappa^\bullet + \tilde{\Psi}_\kappa,$$

so that the remainder $\tilde{\Psi}_\kappa$ satisfies

$$(1 - \Delta)\tilde{\Psi}_\kappa = \lambda (\Psi_\kappa^3 + 3\Psi_\kappa^2 \mathfrak{I}_\kappa + 3\Psi_\kappa \mathfrak{V}_\kappa^\bullet + c_\kappa^{(2)}(\mathfrak{I}_\kappa + \Psi_\kappa)), \quad (7.2)$$

with

$$\Psi_\kappa = \lambda \mathfrak{V}_\kappa^\bullet + \tilde{\Psi}_\kappa.$$

We have

$$\mathfrak{I}_0 \in \mathcal{C}^{-1/2-\varepsilon}, \quad \mathfrak{V}_0^\bullet \in \mathcal{C}^{-1-2\varepsilon}, \quad \mathfrak{V}_0^\bullet \in \mathcal{C}^{-3/2-3\varepsilon}, \quad \mathfrak{V}_0^\bullet \in \mathcal{C}^{1/2-3\varepsilon}.$$

As explained previously, equation (7.2) becomes singular as $\kappa \searrow 0$. The difficulty stems from the presence of the term $\mathfrak{V}_\kappa^\bullet$ on the right-hand side. At best one expects $\tilde{\Psi}_0 \in \mathcal{C}^{1-2\varepsilon}$. However, in this regime the product $\tilde{\Psi}_0 \mathfrak{V}_0^\bullet$ cannot be defined directly, since the sum of regularities is non-positive.

To make sense of this product, we decompose it into paraproducts and a resonant term:

$$(1 - \Delta)\tilde{\Psi}_\kappa = \lambda \left(\Psi_\kappa^3 + 3\Psi_\kappa^2 \mathfrak{I}_\kappa + 3\Psi_\kappa \prec \mathfrak{V}_\kappa^\bullet + 3\Psi_\kappa \succ \mathfrak{V}_\kappa^\bullet + 3\Psi_\kappa \odot \mathfrak{V}_\kappa^\bullet + c_\kappa^{(2)}(\mathfrak{I}_\kappa + \Psi_\kappa) \right). \quad (7.3)$$

From the expected regularities, we anticipate that

$$\Psi_\kappa \in \mathcal{C}^{1/2-3\varepsilon}, \quad \Psi_\kappa \prec \mathfrak{V}_\kappa^\bullet \in \mathcal{C}^{-1-2\varepsilon}, \quad \Psi_\kappa \succ \mathfrak{V}_\kappa^\bullet \in \mathcal{C}^{-1/2-5\varepsilon},$$

uniformly in $\kappa \in (0, 1]$. The resonant product $\Psi_\kappa \odot \mathfrak{V}_\kappa^\bullet$, however, is ill-defined in the limit $\kappa \searrow 0$, since the regularities of the two factors do not add up to a positive number. The key

observation is that this singular term can be combined with the renormalisation contribution $c_\kappa^{(2)}(\mathfrak{l}_\kappa + \Psi_\kappa)$. One then hopes to control the combination

$$3\Psi_\kappa \odot \mathbb{V}_\kappa + c_\kappa^{(2)}(\mathfrak{l}_\kappa + \Psi_\kappa) \in \mathcal{C}^{-1/2-5\varepsilon}.$$

Consequently, the most singular contribution on the right-hand side of (7.3) is the para-product term $\Psi_\kappa \prec \mathbb{V}_\kappa$. This motivates the introduction of the modified unknown

$$\hat{\Psi}_\kappa := \Psi_\kappa - \lambda \mathbb{Y}_\kappa - 3\lambda \Psi_\kappa \prec \mathbb{Y}_\kappa,$$

where

$$\mathbb{Y}_\kappa := G * \mathbb{V}_\kappa,$$

which subtracts precisely the leading singular component.

A direct computation, using in particular the Leibniz rule for paraproducts,

$$\nabla(u \prec v) = (\nabla u) \prec v + u \prec (\nabla v),$$

shows that

$$(1 - \Delta)(\Psi_\kappa \prec \mathbb{Y}_\kappa) - \Psi_\kappa \prec \mathbb{V}_\kappa = -2\nabla\Psi_\kappa \prec \nabla\mathbb{Y}_\kappa - \Delta\Psi_\kappa \prec \mathbb{Y}_\kappa.$$

Hence, $\hat{\Psi}_\kappa$ satisfies

$$\begin{aligned} (1 - \Delta)\hat{\Psi}_\kappa &= \lambda \left(\Psi_\kappa^3 + 3\Psi_\kappa^2 \mathfrak{l}_\kappa + 3\Psi_\kappa \succ \mathbb{V}_\kappa + 3\Psi_\kappa \odot \mathbb{V}_\kappa + c_\kappa^{(2)}(\mathfrak{l}_\kappa + \Psi_\kappa) \right. \\ &\quad \left. - 2\nabla\Psi_\kappa \prec \nabla\mathbb{Y}_\kappa - \Delta\Psi_\kappa \prec \mathbb{Y}_\kappa \right). \end{aligned}$$

The overall strategy can now be formulated precisely. The goal is to prove that

- terms on the right-hand side of the equation for $\tilde{\Psi}_\kappa$ can be controlled in $\mathcal{C}^{-1-2\varepsilon}$ and
- terms on the right-hand side of the equation for $\hat{\Psi}_\kappa$ can be controlled in $\mathcal{C}^{-1/2-5\varepsilon}$

provided $\tilde{\Psi}_\kappa \in \mathcal{C}^{1-2\varepsilon}$ and $\hat{\Psi}_\kappa \in \mathcal{C}^{3/2-5\varepsilon}$. Since the inverse of $1 - \Delta$ gains two orders of regularity, these bounds are stable under the fixed-point map and therefore close the argument in a self-consistent manner. This mechanism forms the analytic backbone of the proof and ultimately enables us to pass to the limit $\kappa \searrow 0$.

Before continuing, we introduce a collection of new stochastic objects. With a suitable choice of counterterms, these objects can be controlled uniformly in the regularisation parameter $\kappa \in (0, 1]$ and one proves existence of the limit $\kappa \searrow 0$.

Definition 7.10. Let $\mathfrak{l}_\kappa, \mathbb{V}_\kappa, \mathbb{V}_\kappa^\bullet, \mathbb{Y}_\kappa$ and $\mathbb{Y}_\kappa^\bullet$ be the stochastic objects introduced in Sec. 6. We define

$$\mathbb{Y}_\kappa^\bullet := \mathbb{Y}_\kappa \odot \mathbb{V}_\kappa + \frac{1}{9}c_\kappa^{(2)}, \quad \mathbb{Y}_\kappa^\bullet := \mathbb{Y}_\kappa^\bullet \odot \mathfrak{l}_\kappa, \quad \mathbb{Y}_\kappa^\bullet := \mathbb{Y}_\kappa^\bullet \odot \mathbb{V}_\kappa + \frac{1}{3}c_\kappa^{(2)}\mathfrak{l}_\kappa.$$

The most problematic term is the resonant product $\Psi_\kappa \odot \mathbb{V}_\kappa$. We rewrite this term so that every ill-defined contribution is absorbed into the enhanced noise. We have

$$\Psi_\kappa \odot \mathbb{V}_\kappa + c_\kappa^{(2)}(\mathbb{I}_\kappa + \Psi_\kappa) = \hat{\Psi}_\kappa \odot \mathbb{V}_\kappa + \lambda \mathbb{Y}_\kappa \odot \mathbb{V}_\kappa + 3\lambda \left(\Psi_\kappa \prec \mathbb{Y}_\kappa \right) \odot \mathbb{V}_\kappa + \frac{1}{3} c_\kappa^{(2)}(\mathbb{I}_\kappa + \Psi_\kappa).$$

The added term $c_\kappa^{(2)}(\mathbb{Y}_\kappa + \Psi_\kappa)$ anticipates the renormalisation needed to compensate for the diverging expectation of the resonant interaction. Using the definition of the commutator we obtain

$$3 \left(\Psi_\kappa \prec \mathbb{Y}_\kappa \right) \odot \mathbb{V}_\kappa + \frac{1}{3} c_\kappa^{(2)} \Psi_\kappa = 3C \left(\Psi_\kappa, \mathbb{Y}_\kappa, \mathbb{V}_\kappa \right) + 3\Psi_\kappa \left(\mathbb{Y}_\kappa \odot \mathbb{V}_\kappa + \frac{1}{9} c_\kappa^{(2)} \right).$$

The point of introducing the commutator is that it allows us to rewrite the problematic resonant product as a resonant product of $\mathbb{Y}_\kappa$ and $\mathbb{V}_\kappa$, which are explicit functions of the noise, thereby facilitating its renormalisation. The second term now involves precisely the renormalised object $\mathbb{Y}_\kappa$, so the singular part has been isolated. Consequently, we arrive at

$$\Psi_\kappa \odot \mathbb{V}_\kappa + c_\kappa^{(2)}(\mathbb{I}_\kappa + \Psi_\kappa) = \hat{\Psi}_\kappa \odot \mathbb{V}_\kappa + \lambda \mathbb{Y}_\kappa + 3C \left(\Psi_\kappa, \mathbb{Y}_\kappa, \mathbb{V}_\kappa \right) + 3\Psi_\kappa \mathbb{Y}_\kappa.$$

At this stage every purely stochastic divergence is encoded in $\mathbb{Y}_\kappa$ and $\mathbb{Y}_\kappa$, which belong to the enhanced noise. All remaining terms are now well-defined using deterministic estimates.

To control the term $\Psi_\kappa^2 \mathbb{I}_\kappa$ we rewrite it as

$$\Psi_\kappa^2 \mathbb{I}_\kappa = \tilde{\Psi}_\kappa^2 \mathbb{I}_\kappa + 2\tilde{\Psi}_\kappa \mathbb{Y}_\kappa \mathbb{I}_\kappa + \mathbb{Y}_\kappa^2 \mathbb{I}_\kappa$$

and use the result of the exercise below to estimate the last, purely stochastic term in terms of the enhanced noise.

Exercise 7.5. *Show that*

$$\|\mathbb{Y}_\kappa^2 \mathbb{I}_\kappa\|_{\mathcal{C}^{-1/2-\varepsilon}} \lesssim \|\mathbb{Y}_\kappa\|_{\mathcal{C}^{-4\varepsilon}} + \|\mathbb{I}_\kappa\|_{\mathcal{C}^{-1/2-\varepsilon}} \|\mathbb{Y}_\kappa\|_{\mathcal{C}^{1/2-3\varepsilon}}$$

and

$$\|\mathbb{Y}_\kappa^2 \mathbb{I}_\kappa\|_{\mathcal{C}^{-1/2-\varepsilon}} \lesssim \|\mathbb{Y}_\kappa\|_{\mathcal{C}^{1/2-3\varepsilon}} \|\mathbb{Y}_\kappa\|_{\mathcal{C}^{-4\varepsilon}} + \|\mathbb{I}_\kappa\|_{\mathcal{C}^{-1/2-\varepsilon}} \|\mathbb{Y}_\kappa\|_{\mathcal{C}^{1/2-3\varepsilon}}^2.$$

Definition 7.11 (Enhanced noise). Let

$$\mathcal{N} := \mathcal{C}^{-1/2-\varepsilon} \times \mathcal{C}^{-1-2\varepsilon} \times \mathcal{C}^{-3/2-3\varepsilon} \times \mathcal{C}^{-4\varepsilon} \times \mathcal{C}^{-4\varepsilon} \times \mathcal{C}^{-1/2-5\varepsilon}.$$

For $\Xi = (\Xi_1, \dots, \Xi_6) \in \mathcal{N}$, we define

$$\|\Xi\|_{\mathcal{N}} := \|\Xi_1\|_{\mathcal{C}^{-1/2-\varepsilon}} + \|\Xi_2\|_{\mathcal{C}^{-1-2\varepsilon}} + \|\Xi_3\|_{\mathcal{C}^{-3/2-3\varepsilon}} + \|\Xi_4\|_{\mathcal{C}^{-4\varepsilon}} + \|\Xi_5\|_{\mathcal{C}^{-4\varepsilon}} + \|\Xi_6\|_{\mathcal{C}^{-1/2-5\varepsilon}}.$$

For $\kappa \in (0, 1]$, we define

$$\Xi_\kappa = \left(\mathbb{I}_\kappa, \mathbb{V}_\kappa, \mathbb{Y}_\kappa, \mathbb{Y}_\kappa, \mathbb{Y}_\kappa, \mathbb{Y}_\kappa, \mathbb{Y}_\kappa \right) \in \mathcal{N}.$$

The space $\mathcal{N}$ collects all the stochastic objects required to formulate the fixed point problem. Each component corresponds to one of the renormalised trees in the expansion. The assigned regularities reflect the scaling heuristics, based on the naive expectation that the appropriately renormalized resonant product of $u \in \mathcal{C}^\alpha$ and $v \in \mathcal{C}^\beta$ can be controlled stochastically in $\mathcal{C}^{\alpha+\beta}$, even when $\alpha + \beta$ is not positive.

Remark 7.12. Note that we have

$$\|\Upsilon_\kappa^\bullet\|_{\mathcal{C}^{1-2\varepsilon}} \lesssim \|\mathbf{V}_\kappa^\bullet\|_{\mathcal{C}^{-1-2\varepsilon}}, \quad \|\Upsilon_\kappa^{\bullet\bullet}\|_{\mathcal{C}^{1/2-3\varepsilon}} \lesssim \|\mathbf{V}_\kappa^{\bullet\bullet}\|_{\mathcal{C}^{-3/2-3\varepsilon}}.$$

Lemma 7.13. *Let*

$$c_\kappa^{(1)} := -3 \mathbb{E} \mathbf{I}_\kappa^2, \quad c_\kappa^{(2)} := -9 \mathbb{E} \Upsilon_\kappa^\bullet \mathbf{V}_\kappa^\bullet.$$

Then the limit

$$\lim_{\kappa \searrow 0} \Xi_\kappa =: \Xi_0 \in \mathcal{N} \tag{7.4}$$

exists almost surely and there exists a random variable $R \in [1, \infty]$ such that $\mathbb{E}R^n < \infty$ for all $n \in \mathbb{N}_0$ and

$$\|\Xi_\kappa\|_{\mathcal{N}} \leq R$$

for all $\kappa \in [0, 1]$.

Proof. See [MWX16] for a proof of a very similar result. $\square$

This is the probabilistic core of the construction. Once the convergence of the enhanced noise is known, the rest of the argument becomes deterministic.

Definition 7.14. The map

$$\mathbf{Q} = (\hat{\mathbf{Q}}, \tilde{\mathbf{Q}}) : \mathcal{C}^{3/2-5\varepsilon} \times \mathcal{C}^{1-2\varepsilon} \times \mathcal{N} \rightarrow \mathcal{C}^{3/2-5\varepsilon} \times \mathcal{C}^{1-2\varepsilon}$$

is defined by

$$\begin{aligned} \hat{\mathbf{Q}}[\hat{\psi}, \tilde{\psi}; \Xi] &:= \lambda G * \left(\psi^3 + 3\tilde{\psi}^2 \mathbf{I} + 6\tilde{\psi} \Upsilon^\bullet \mathbf{I} + 3\Upsilon^{\bullet 2} \mathbf{I} + 3\psi \succ \mathbf{V}^\bullet + 3\hat{\psi} \odot \mathbf{V}^\bullet + 3\lambda \Upsilon^{\bullet\bullet} \right. \\ &\quad \left. + 9C(\psi, \Upsilon^\bullet, \mathbf{V}^\bullet) + 9\psi \Upsilon^{\bullet\bullet} - 2\nabla\psi \prec \nabla \Upsilon^\bullet - \Delta\psi \prec \Upsilon^\bullet \right), \end{aligned}$$

and

$$\tilde{\mathbf{Q}}[\hat{\psi}, \tilde{\psi}; \Xi] := \hat{\mathbf{Q}}[\psi, \hat{\psi}; \Xi] + \lambda G * \left(3\psi \prec \mathbf{V}^\bullet + 2\nabla\psi \prec \nabla \Upsilon^\bullet + \Delta\psi \prec \Upsilon^\bullet \right),$$

with

$$\psi = \lambda \mathbf{V}^\bullet + \tilde{\psi}$$

and

$$\Xi = \left(\mathbf{I}, \mathbf{V}^\bullet, \Upsilon^\bullet, \Upsilon^{\bullet\bullet}, \Upsilon^{\bullet\bullet}, \Upsilon^{\bullet\bullet} \right) \in \mathcal{N}.$$

The map $\mathbf{Q}$ encodes the mild formulation of the equation in paracontrolled form. All nonlinear interactions have been rewritten so that only admissible products appear. The convolution with G provides two derivatives of smoothing, which compensates for the low regularity of the forcing terms. The lemma below verifies that the paracontrolled ansatz is consistent and the fixed point problem is not artificial — it reconstructs an actual solution of the equation.

Lemma 7.15. *Let $\varepsilon \in (0, 1]$. Suppose that $(\hat{\Psi}_\kappa, \tilde{\Psi}_\kappa) \in \mathcal{C}^{3/2-5\varepsilon} \times \mathcal{C}^{1-2\varepsilon}$ solves the fixed point problem*

$$(\hat{\Psi}_\kappa, \tilde{\Psi}_\kappa) = \mathbf{Q}[\hat{\Psi}_\kappa, \tilde{\Psi}_\kappa; \Xi_\kappa]. \quad (7.5)$$

Then $\Phi_\kappa = \mathfrak{I}_\kappa + \lambda \mathfrak{Y}_\kappa + \tilde{\Psi}_\kappa$ is a mild solution of the original SPDE (7.1).

The next lemma is the core of the deterministic part of the argument. It says that for small λ (depending polynomially on the size of the enhanced noise) the map $\mathbf{Q}_\kappa$ is a contraction on a suitable ball.

Lemma 7.16. *For $R \in [1, \infty)$ let $\mathcal{B}_R$ be the closed ball in $\mathcal{C}^{3/2-5\varepsilon} \times \mathcal{C}^{1-2\varepsilon}$ of radius R . For all $\kappa \in [0, 1]$ let $\mathbf{Q}_\kappa[\hat{\psi}, \tilde{\psi}] := \mathbf{Q}[\hat{\psi}, \tilde{\psi}; \Xi_\kappa]$. Assume that (7.4) and (7.4) hold true. Then there exists $c > 0$ and $p \in \mathbb{N}_+$ such that for all $\lambda \in [-\lambda_\star, \lambda_\star]$ with $\lambda_\star = cR^{-p}$, and $\kappa \in [0, 1]$ the map $\mathbf{Q}_\kappa : \mathcal{B}_R \rightarrow \mathcal{B}_R$ is well defined and is a contraction with the Lipschitz constant less than $1/2$. Moreover, $\lim_{\kappa \searrow 0} \mathbf{Q}_\kappa[\psi] = \mathbf{Q}_0[\psi]$ for all $\psi \in \mathcal{B}_R$.*

Exercise 7.6. *Prove the deterministic lemma stated above and combine it with the two previous lemmas to complete the proof of Proposition 7.1.*

Proposition 7.17. *The statement of Theorem 5.1 holds true for $\sigma \in (7d/18, 2d/5]$.*

Exercise 7.7. *Proof the above proposition by showing that the method of the proof presented in this section generalizes to $\sigma \in (7d/18, 2d/5]$. To this end, define $\mathbf{Q}$ as above and check that*

$$\mathbf{Q} : \mathcal{C}^{5\alpha+2\sigma} \times \mathcal{C}^{2\alpha+\sigma} \times \mathcal{N} \rightarrow \mathcal{C}^{5\alpha+2\sigma} \times \mathcal{C}^{2\alpha+\sigma}.$$

You can assume that for all $\alpha < 0$ and $\beta \in \mathbb{R}$, we have

$$\|f \prec (G * g) - G * (f \prec g)\|_{\mathcal{C}^{\alpha+\beta+\sigma+1}} \lesssim \|\nabla f\|_{\mathcal{C}^\alpha} \|g\|_{\mathcal{C}^\beta},$$

where G is the Green function for $(1 - \Delta)^{\sigma/2}$.

To reach the more singular regime $\sigma \leq 7d/18$, a more sophisticated framework is required. One possible strategy would be to further develop the paracontrolled approach by exploiting the iterated commutator techniques introduced in [BB19]. However, it is arguably more convenient to rely on alternative frameworks. Two natural options are the theory of regularity structures, originally introduced by Hairer in [Hai14], and the flow equation approach proposed in [Duc25a].

8 Regularity structure approach

9 Flow equation approach

9.1 Effective equation

The starting point of the flow equation approach is a reformulation of the original SPDE (5.2) in terms of an *effective equation*. This provides an equivalent description of the dynamics in which the nonlinear interactions are tracked under progressive coarse-graining. A key advantage of this formulation is that, unlike the original equation, it remains meaningful in the limit in which the regularization is removed.

Definition 9.1. We use the notation

$$\mathbb{M} := \mathbb{R}^d, \quad \mathbb{T} := (\mathbb{R}/2\pi\mathbb{Z})^d, \quad \|\bullet\| := \|\bullet\|_{L^\infty(\mathbb{M})}$$

and identify function $\mathbb{T} \rightarrow \mathbb{R}$ with periodic functions $\mathbb{M} \rightarrow \mathbb{R}$.

Recall that our goal is to construct a solution Φ_κ of the equation

$$\Phi_\kappa = G * F_\kappa[\Phi_\kappa], \quad F_\kappa[\varphi] := \xi_\kappa + \lambda \varphi^3 + \sum_{i=1}^{i_\sharp} \lambda^i c_\kappa^{(i)} \varphi, \quad (9.1)$$

and to prove that, for a suitable choice of counterterms $(c_\kappa^{(i)})_{i=1, \dots, i_\sharp}$, the limit

$$\lim_{\kappa \searrow 0} \Phi_\kappa =: \Phi_0 \in \mathcal{S}'(\mathbb{M})$$

exists. The functional F_κ will be referred to as the *force* and $G \in L^1(\mathbb{M})$ denotes the fundamental solution of the differential operator $(1 - \Delta)^{\sigma/2}$.

In this section we introduce a system of equations with the following properties:

- For every $\kappa \in (0, 1]$ there is a correspondence between solutions of this system and solutions of (9.1).
- In contrast to (9.1) the system remains well posed in the limit $\kappa \searrow 0$.

Thus, this system of equations plays an analogous role to (6.2) or (6.5) in the Da Prato-Debussche approach and (7.5) in the paracontrolled approach. Its advantage is that it remains applicable in the entire subcritical regime.

To this end, we introduce a *scale decomposition* of the Green function G together with the notion of an *effective force*. A scale decomposition of G is a family of kernels $G_\mu \in L^1(\mathbb{M})$ parametrized by $\mu \in [0, 1]$ such that

- (1) $G_0 = G$ and $G_1 = 0$,

(3) the map

$$[0, 1] \ni \mu \mapsto G_\mu \in L^1(\mathbb{M})$$

is continuous and piecewise continuously differentiable,

(3) for $\mu \in (0, 1]$, $\dot{G}_\mu := \partial_\mu G_\mu \in C^\infty(\mathbb{M})$ “lives” at spatial scales μ , that is for all multi-indices $a, b \in \mathbb{N}_0^d$, the bounds

$$\|x^a \partial^b \dot{G}_\mu\|_{L^\infty(\mathbb{M})} \lesssim \mu^{\sigma-d-1+|a|-|b|}, \quad \|x^a \partial^b \dot{G}_\mu\|_{L^1(\mathbb{M})} \lesssim \mu^{\sigma-1+|a|-|b|} \quad (9.2)$$

hold uniformly over $\mu \in (0, 1]$.

Exercise 9.1. Fix $\chi \in C^\infty(\mathbb{R}^d)$ such that $\chi(x) = 0$ for $|x| \leq 1/4$ and $\chi(x) = 1$ for $|x| \geq 1/2$ and let $\chi_\mu(x) := \chi(x(1-\mu)/\mu)$ for $\mu \in (0, 1]$. Show that the following scale decompositions of the Green function

$$(a) \quad G_\mu := \chi_\mu G,$$

$$(b) \quad G := \mathcal{F}^{-1}(\chi_\mu(1/\bullet) \mathcal{F}G),$$

satisfy the bounds (9.2).

Given the force functional F_κ , an *effective force* is a family of functionals

$$F_{\kappa, \mu} : C(\mathbb{T}) \rightarrow C(\mathbb{T}), \quad \mu \in [0, 1],$$

such that $F_{\kappa, 0} = F_\kappa$ and the map

$$[0, 1] \ni \mu \mapsto F_{\kappa, \mu}[\varphi] \in C(\mathbb{T})$$

is continuous and piecewise continuously differentiable for all $\varphi \in C(\mathbb{T})$. Moreover, we assume that $F_{\kappa, \mu}$ is of polynomial type, that is for every $\varphi, \psi \in C(\mathbb{T})$ the directional derivatives

$$D^k V[\varphi] \cdot \psi^{\otimes k} := \partial_\tau^k V[\varphi + \tau \psi] \Big|_{\tau=0}$$

exist for all $k \in \mathbb{N}_+$ and are nonzero for only finitely many $k \in \mathbb{N}_+$.

Suppose that Φ_κ solves the original equation

$$\Phi_\kappa = G * F_\kappa[\Phi_\kappa].$$

For $\mu \in [0, 1]$ we define

$$\Phi_{\kappa, \mu} = G_\mu * F_\kappa[\Phi_\kappa].$$

The family $(\Phi_{\kappa, \mu})_{\mu \in (0, 1]}$ will be called the *coarse-grained process*. By construction,

$$\Phi_{\kappa, 0} = \Phi_\kappa, \quad \Phi_{\kappa, 1} = 0.$$

For $\mu \in [0, 1]$ we define the *remainder* $\zeta_{\kappa,\mu}$ through

$$F_\kappa[\Phi_\kappa] = F_{\kappa,\mu}[\Phi_{\kappa,\mu}] + \zeta_{\kappa,\mu}. \quad (9.3)$$

Since $F_{\kappa,0} = F_\kappa$ and $\Phi_{\kappa,0} = \Phi_\kappa$, it follows that $\zeta_{\kappa,0} = 0$. Because the left-hand side of (9.3) does not depend on μ , we obtain

$$\partial_\mu \Phi_{\kappa,\mu} = \dot{G}_\mu * (F_{\kappa,\mu}[\Phi_{\kappa,\mu}] + \zeta_{\kappa,\mu}) \quad (9.4)$$

and

$$\partial_\mu \zeta_{\kappa,\mu} = -(\partial_\mu F_{\kappa,\mu})[\Phi_{\kappa,\mu}] - DF_{\kappa,\mu}[\Phi_{\kappa,\mu}] \cdot \partial_\mu \Phi_{\kappa,\mu}. \quad (9.5)$$

Substituting (9.4) into (9.5) yields

$$\partial_\mu \zeta_{\kappa,\mu} = -(\partial_\mu F_{\kappa,\mu})[\Phi_{\kappa,\mu}] - DF_{\kappa,\mu}[\Phi_{\kappa,\mu}] \cdot (\dot{G}_\mu * (F_{\kappa,\mu}[\Phi_{\kappa,\mu}] + \zeta_{\kappa,\mu})). \quad (9.6)$$

Introducing the notation

$$H_{\kappa,\mu}[\varphi] := \partial_\mu F_{\kappa,\mu}[\varphi] + DF_{\kappa,\mu}[\varphi] \cdot (\dot{G}_\mu * F_{\kappa,\mu}[\varphi]), \quad (9.7)$$

we can rewrite (9.6) as

$$\partial_\mu \zeta_{\kappa,\mu} = -H_{\kappa,\mu}[\Phi_{\kappa,\mu}] - DF_{\kappa,\mu}[\Phi_{\kappa,\mu}] \cdot (\dot{G}_\mu * \zeta_{\kappa,\mu}). \quad (9.8)$$

Combining (9.4) and (9.8) with the boundary conditions

$$\Phi_{\kappa,1} = 0, \quad \zeta_{\kappa,0} = 0,$$

we arrive at the following system:

$$\begin{cases} \Phi_{\kappa,\mu} = - \int_\mu^1 \dot{G}_\eta * (F_{\kappa,\eta}[\Phi_{\kappa,\eta}] + \zeta_{\kappa,\eta}) \, d\eta, \\ \zeta_{\kappa,\mu} = - \int_0^\mu \left(H_{\kappa,\eta}[\Phi_{\kappa,\eta}] + DF_{\kappa,\eta}[\Phi_{\kappa,\eta}] \cdot (\dot{G}_\eta * \zeta_{\kappa,\eta}) \right) \, d\eta. \end{cases} \quad (9.9)$$

The system (9.9) for $(\Phi_{\kappa,\bullet}, \zeta_{\kappa,\bullet})$ is called the *effective equation*. Formally, there is a one-to-one correspondence between solutions of the original equation (9.1) and solutions of (9.9). Indeed, given a solution Φ_κ of the original equation, one constructs a pair $(\Phi_{\kappa,\bullet}, \zeta_{\kappa,\bullet})$ as above with $H_{\kappa,\bullet}$ defined by (9.7). Conversely, if $(\Phi_{\kappa,\bullet}, \zeta_{\kappa,\bullet})$ solves (9.9), then (9.3) holds and $\Phi_\kappa = \Phi_{\kappa,0}$ solves the original equation. The advantage of the formulation (9.9) is that, as we will see, for a suitable choice of the effective force $F_{\kappa,\bullet}$, the system remains meaningful in the limit $\kappa \searrow 0$.

The coarse-grained process $\Phi_{\kappa,\mu}$ lives at spatial scales of order μ . In particular, we expect that

$$\|\partial^a \Phi_{\kappa,\mu}\| \lesssim \mu^{\alpha-|a|}$$

uniformly over $\kappa, \mu \in (0, 1]$, where $\alpha = \sigma - d/2 - \varepsilon < 0$ denotes the expected regularity of the solution to the original equation. It is therefore natural to control the coarse-grained process using the norm

$$\|\Phi_{\kappa, \bullet}\| := \sup_{\mu \in (0, 1]} \mu^{-\alpha} \|\mathbf{P}_\mu \Phi_{\kappa, \mu}\|,$$

where $\mathbf{P}_\mu := (1 - \mu^2 \Delta)^p$ and $p \in \mathbb{N}_+$ is fixed and sufficiently large. It is then convenient to introduce a regularizing kernel K_μ , defined as the fundamental solution of $\mathbf{P}_\mu$. With this notation we obtain, for instance,

$$\dot{G}_\eta * F_{\kappa, \eta}[\varphi] = \mathbf{P}_\eta \dot{G}_\eta * K_\eta * F_{\kappa, \eta}[K_\eta * \mathbf{P}_\eta \varphi].$$

Since the effective force involves the noise ξ and

$$\|K_\eta * \xi\| \lesssim \mu^{\alpha - \sigma},$$

we expect that

$$\|K_\eta * F_{\kappa, \eta}[K_\eta * \varphi_\eta]\| \lesssim \mu^{\alpha - \sigma}, \quad (9.10)$$

uniformly over $\kappa, \mu \in (0, 1]$ and over all $\varphi_\bullet : (0, 1] \rightarrow L^\infty$ such that $\|\varphi_\bullet\| \lesssim 1$. Note that $\alpha - \sigma$ coincides with the regularity of the noise. Suppose further that

$$\|K_\eta * H_{\kappa, \eta}[K_\eta * \varphi_\eta]\| \lesssim \eta^{\beta - 1} \quad (9.11)$$

for some β . Since the second equation in (9.9) involves integration over η from 0 to μ , we must require $\beta > 0$ in order for $\eta^{\beta - 1}$ to be integrable. It is therefore natural to control the remainder $\zeta_{\kappa, \mu}$ using the norm

$$\|\zeta_{\kappa, \bullet}\| := \sup_{\mu \in (0, 1]} \mu^{-\beta} \|K_\mu * \zeta_{\kappa, \mu}\|.$$

A more detailed analysis shows that if $F_{\kappa, \bullet}$ and $H_{\kappa, \bullet}$ satisfy (9.10) with $\alpha = \sigma - d/2 - \varepsilon$ and (9.11) with $\beta > 0$, and if in addition $DF_{\kappa, \bullet}$ satisfies a bound of a similar type, then (9.9) can be reformulated as a fixed-point problem for a suitable map. This map is a contraction on the ball

$$\|\Phi_{\kappa, \bullet}\| \vee \|\zeta_{\kappa, \bullet}\| \lesssim 1$$

for all $|\lambda| \leq \lambda_\star$, provided that $\lambda_\star > 0$ is chosen sufficiently small. We emphasize that the assumption $\beta > 0$ in (9.11) is essential. One can expect this condition to hold only if the effective force $F_{\kappa, \bullet}$ is chosen so that certain cancellations occur on the right-hand side of (9.7). A natural possibility is to consider an effective force satisfying the flow equation

$$\partial_\mu F_{\kappa, \mu}[\varphi] + DF_{\kappa, \mu}[\varphi] \cdot (\dot{G}_\mu * F_{\kappa, \mu}[\varphi]) = 0. \quad (9.12)$$

In this case one has $H_{\kappa, \bullet} = 0$ and consequently $\zeta_{\kappa, \bullet} = 0$, so that the system of equations (9.9) reduces to

$$\Phi_{\kappa, \mu} = - \int_\mu^1 \dot{G}_\eta * F_{\kappa, \eta}[\Phi_{\kappa, \eta}] d\eta.$$

In practice, however, it is often more convenient to work with an approximate solution of the flow equation (9.12) and to analyze the full system (9.9).

9.2 Construction of effective force

The construction of the effective force begins with an appropriate ansatz. Since the non-linearity in (9.1) is polynomial, it is natural to assume that the effective force itself has a polynomial structure. Moreover, by subcriticality, higher-order contributions are expected to exhibit improved behaviour. To exploit this feature, we introduce a grading in powers of the coupling constant λ . This leads to the following ansatz

$$\langle F_{\kappa,\mu}[\varphi], \psi \rangle := \sum_{i=0}^{i_b} \sum_{m=0}^{3i} \lambda^i \langle F_{\kappa,\mu}^{i,m}, \psi \otimes \varphi^{\otimes m} \rangle \quad (9.13)$$

for all $\psi, \varphi \in \mathcal{S}(\mathbb{M})$. The parameter $i_b \in \mathbb{N}_+$ will be chosen big enough. Note that the constraint $m \leq 3i$ reflects the cubic nonlinearity. The distributions $F_{\kappa,\mu}^{i,m} \in \mathcal{S}'(\mathbb{M}^{1+m})$ that appear on the right-hand side of the above equality are called the effective force kernels. By definition $F_{\kappa,\mu}^{i,m} \in \mathcal{S}'(\mathbb{M}^{1+m})$ are such that the expression $\langle F_{\kappa,\mu}^{i,m}, \psi \otimes \varphi_1 \otimes \dots \otimes \varphi_m \rangle$ is invariant under permutations of the test functions $\varphi_1, \dots, \varphi_m \in \mathcal{S}(\mathbb{M})$. The kernels $F_{\kappa}^{i,m}$ of the force F_{κ} are defined by an equality analogous to (9.13).

Remark 9.2. Let us list the non-vanishing force kernels $F_{\kappa}^{i,m}$:

$$\begin{aligned} F_{\kappa}^{0,0}(x) &= \xi_{\kappa}(x), & F_{\kappa}^{1,3}(x; dy_1, dy_2, dy_3) &= \delta_x(dy_1) \delta_x(dy_2) \delta_x(dy_3), \\ F_{\kappa}^{i,1}(x; dy_1) &= c_{\kappa}^{(i)} \delta_x(dy_1), & i &\in \{0, \dots, i_{\#}\}, \end{aligned}$$

where $(c_{\kappa}^{(i)})_{i \in \{1, \dots, i_{\#}\}}$ are the so-called counterterms. It is easy to see that $F_{\kappa}^{0,0} \in \mathcal{V}^0$, $F_{\kappa}^{1,3} \in \mathcal{V}^3$, $F_{\kappa}^{i,1} \in \mathcal{V}^1$, where the spaces $\mathcal{V}^m$ are introduced in the definition below.

Definition 9.3. For $m \in \mathbb{N}_0$ the space $\mathcal{V}^m$ consists of maps $V : \mathbb{M} \times \text{Borel}(\mathbb{M}^m) \rightarrow \mathbb{R}$ satisfying the following conditions:

- (1) for every $A \in \text{Borel}(\mathbb{M}^m)$ the map $x \mapsto V(x; A + x)$ is continuous and 2π periodic, where $A + x := \{(y_1 + x, \dots, y_m + x) \in \mathbb{M}^m \mid (y_1, \dots, y_m) \in A\}$ for $A \subset \mathbb{M}^m$,
- (2) for every $x \in \mathbb{M}$ the map $A \mapsto V(x; A)$ is a measure with finite total variation,
- (3) the following norm

$$\|V\|_{\mathcal{V}^m} := \sup_{x \in \mathbb{M}} \int_{\mathbb{M}^m} |V(x; dy_1 \dots dy_m)|$$

is finite.

Recall that our goal is to construct an effective force that approximately solves the flow equation (9.12), with a small error term as defined in (9.7). More precisely, we require that

$$H_{\kappa,\mu} = O(\lambda^{i_b+1}).$$

We will see that if $i_b \in \mathbb{N}_+$ is chosen sufficiently large, then the bound (9.11) is automatically satisfied for some $\beta > 0$. It follows from (9.7) that

$$\partial_\mu F_{\kappa,\mu}[\varphi] + \mathbf{D}F_{\kappa,\mu}[\varphi] \cdot (\dot{G}_\mu * F_{\kappa,\mu}[\varphi]) = H_{\kappa,\mu}[\varphi] = O(\lambda^{i_b+1}).$$

As a result, the effective force kernels $(F_{\kappa,\bullet}^{i,m})_{i \in \{0, \dots, i_b\}, m \in \mathbb{N}_0}$ satisfy the following flow equation

$$\partial_\mu F_{\kappa,\mu}^{i,m} = - \sum_{j=0}^i \sum_{k=0}^m (1+k) \mathbf{B}(\dot{G}_\mu, F_{\kappa,\mu}^{j,1+k}, F_{\kappa,\mu}^{i-j,m-k}), \quad (9.14)$$

where the map $\mathbf{B}$ is introduced below.

Definition 9.4. Let $m \in \mathbb{N}_0$, $k \in \{0, \dots, m\}$. The map $\mathbf{B} : \mathcal{S}(\mathbb{M}) \times \mathcal{V}^{1+k} \times \mathcal{V}^{m-k} \rightarrow \mathcal{V}^m$ is defined by

$$\begin{aligned} \mathbf{B}(G, W, U)(x; dy_1, \dots, dy_m) \\ := \frac{1}{m!} \sum_{\pi \in \mathcal{P}_m} \int_{\mathbb{M}^2} W(x; dy, dy_{\pi(1)}, \dots, dy_{\pi(k)}) G(y-z) U(z; dy_{\pi(k+1)}, \dots, dy_{\pi(m)}) dz. \end{aligned}$$

Exercise 9.2. Prove that the map $\mathbf{B} : \mathcal{S}(\mathbb{M}) \times \mathcal{V}^{1+k} \times \mathcal{V}^{m-k} \rightarrow \mathcal{V}^m$ is well defined and

$$\|\mathbf{B}(G, W, U)\|_{\mathcal{V}^m} \leq \|G\|_{L^1} \|W\|_{\mathcal{V}^{1+k}} \|U\|_{\mathcal{V}^{m-k}}.$$

The basic idea behind the flow equation approach is a recursive construction of the effective force kernels $(F_{\kappa,\mu}^{i,m})_{i \in \{0, \dots, i_b\}, m \in \mathbb{N}_0}$ for all $\kappa \in (0, 1]$, $\mu \in [0, 1]$. For each $\kappa \in (0, 1]$ we define the kernels in such a way that the map

$$[0, 1] \ni \mu \mapsto F_{\kappa,\mu}^{i,m} \in \mathcal{V}^m$$

is continuous and continuously differentiable for $\mu \in (0, 1]$ using the following recursive algorithm:

- (0) We set $F_{\kappa,\mu}^{0,0} = \xi_\kappa$ and $F_{\kappa,\mu}^{i,m} = 0$ if $m > 3i$,
- (I) Assuming that all $F_{\kappa,\mu}^{i,m}$ with $i < i_\circ$, or $i = i_\circ$ and $m > m_\circ$ were constructed we define $\dot{F}_{\kappa,\mu}^{i,m}$ with $i = i_\circ$ and $m = m_\circ$ to be the right-hand side of (9.14).
- (II) Subsequently, $F_{\kappa,\mu}^{i,m}$ is defined by $F_{\kappa,\mu}^{i,m} = F_\kappa^{i,m} + \int_0^\mu \dot{F}_{\kappa,\eta}^{i,m} d\eta$.

Definition 9.5. The finite list of the effective force kernels $(F_{\kappa,\bullet}^{i,m})_{i \in \{0, \dots, i_b\}, m \in \{0, \dots, 3i\}}$ is called the *enhanced noise*.

Remark 9.6. The above procedure cannot be used to construct directly $F_{\kappa,\mu}^{i,m}$ with $\kappa = 0$. In fact, we expect that $F_{\kappa,\mu}^{i,m} \notin \mathcal{V}^m$. For example, $F_{0,\mu}^{0,0} = \xi \notin C(\mathbb{T}) = \mathcal{V}^0$. Instead, the kernels $F_{0,\mu}^{i,m}$ are defined probabilistically. The stochastic estimates for the enhanced noise are stated in Theorem 9.10 below.

Remark 9.7. Note that the effective force kernels depend implicitly on the counterterms.

Exercise 9.3. Check that the condition $F_{\kappa,\mu}^{i,m} = 0$ if $m > 3i$ is consistent with the conditions stated in Items (I) and (II). Prove that $F_{\kappa,\mu}^{i,m} = 0$ for all $m > (2i + 1) \wedge 3i$.

Remark 9.8. Let us list some examples of effective force kernels. To this end, it will be convenient to use a diagrammatical notation. We view the diagrams as placeholders for certain multi-linear functionals of the noise. Since the diagrams do not play any role in the flow equation approach we refrain from defining precise rules that are used to draw them. Instead, for each diagram we provided an explicit expression it represents. Note that the edges of the diagrams that are introduced below represent the fluctuation propagator $(G - G_\mu)(x - y) =: x \text{---} y$ and not the Green function $G(x - y) =: x \text{---} y$. First note that

$$F_{\kappa,\mu}^{0,0}(x) = \xi_\kappa(x) =: \bullet_\kappa(x),$$

$$F_{\kappa,\mu}^{1,3}(x; dy_1, dy_2, dy_3) = F_\kappa^{1,3}(x; dy_1, dy_2, dy_3) =: \circ(x; dy_1, dy_2, dy_3).$$

Let

$$\mathfrak{I}_{\kappa,\mu}(x) := ((G - G_\mu) * \bullet_\kappa)(x), \quad \mathfrak{V}_{\kappa,\mu}(x) := (\mathfrak{I}_{\kappa,\mu}(x))^2 + c_\kappa^{(1)}/3.$$

We have

$$F_{\kappa,\mu}^{1,2}(x; dy_1, dy_2) = 3\mathfrak{I}_{\kappa,\mu}(x) \delta_x(dy_1) \delta_x(dy_2) =: 3\mathfrak{J}_{\kappa,\mu}(x; dy_1, dy_2),$$

$$F_{\kappa,\mu}^{1,1}(x; dy_1) = 3\mathfrak{V}_{\kappa,\mu}(x) \delta_x(dy_1) =: 3\mathfrak{Y}_{\kappa,\mu}(x; dy_1),$$

$$F_{\kappa,\mu}^{1,0}(x) = (\mathfrak{I}_{\kappa,\mu}(x))^3 + c_\kappa^{(1)} \mathfrak{I}_{\kappa,\mu}(x) =: \mathfrak{Z}_{\kappa,\mu}(x),$$

The kernel

$$F_{\kappa,\mu}^{2,5}(x; dy_1, \dots, dy_5) =: 3\mathfrak{G}_\mu(x; dy_1, \dots, dy_5)$$

coincides with the symmetric part of

$$3\delta_x(dy_1) \delta_x(dy_2) (G - G_\mu)(x - y_3) \delta_{y_3}(dy_4) \delta_{y_3}(dy_5) dy_3.$$

The construction of kernels $F_{\kappa,\mu}^{2,4}$, $F_{\kappa,\mu}^{2,3}$, $F_{\kappa,\mu}^{2,2}$ is left as an exercise. Let

$$\mathfrak{Y}_{\kappa,\mu}(x) := ((G - G_\mu) * \mathfrak{V}_{\kappa,\mu})(x).$$

The remaining second order kernels are given by

$$F_{\kappa,\mu}^{2,1}(x; dy_1) = 9\mathfrak{V}_{\kappa,\mu}(x) (G - G_\mu)(x - y_1) \mathfrak{V}_{\kappa,\mu}(y_1) dy_1 + c_\kappa^{(2)} \delta_x(dy_1)$$

$$+ 6\mathfrak{I}_{\kappa,\mu}(x) \mathfrak{Y}_{\kappa,\mu}(x) \delta_x(dy_1) =: 9\mathfrak{W}_{\kappa,\mu}(x; dy_1) + 6\mathfrak{Y}_{\kappa,\mu}(x; dy_1)$$

and

$$F_{\kappa,\mu}^{2,0}(x) = 3\mathfrak{V}_{\kappa,\mu}(x) \mathfrak{Y}_{\kappa,\mu}(x) + c_\kappa^{(2)} \mathfrak{I}_{\kappa,\mu}(x) =: 3\mathfrak{X}_{\kappa,\mu}(x).$$

Exercise 9.4. Convince yourself that the expression given in Remark 9.8 satisfy the condition $F_{\kappa,0}^{i,m} = F_\kappa^{i,m}$, where $F_\kappa^{i,m}$ are the force kernels listed in Remark 9.2.

Exercise 9.5. Using the notation $\dot{G}_\mu(x-y) := x \text{---} y$ draw the diagrams representing the kernels $\partial_\mu F_{\kappa,\mu}^{i,m}$ for $i \in \{0,1,2\}$. Verify the formulas given in Remark 9.8 and write explicit expressions for the kernels $F_{\kappa,\mu}^{2,4}, F_{\kappa,\mu}^{2,3}, F_{\kappa,\mu}^{2,2}$. For simplicity, you can ignore numerical prefactors.

9.3 Stochastic estimates

In this section we formulate the stochastic estimates for the effective force kernels that allow to close the fixed-point estimates for the effective equation (9.9). We first introduce a convenient notation that encodes the scaling of the effective force kernels appearing in the expansion. The quantity $\varrho_\varepsilon(i, m)$ defined below measures the expected homogeneity in the scale parameter μ of the kernel $F_{\kappa,\mu}^{i,m}$ after convolution with the regularizing kernels K_μ .

Definition 9.9. Let $\varepsilon \geq 0$ and $\alpha \equiv \alpha_\varepsilon := \sigma - d/2 - \varepsilon$, $\gamma \equiv \gamma_\varepsilon := 3\sigma - d - 3\varepsilon$. For $i, m \in \mathbb{N}_0$, we define

$$\varrho_\varepsilon(i, m) := \alpha_\varepsilon - \sigma - m\alpha_\varepsilon + i\gamma_\varepsilon \in \mathbb{R}.$$

We omit ε if $\varepsilon = 0$. Let $i_b, i_\sharp \in \mathbb{N}_+$ be the smallest positive integers such that $\varrho(i_b + 1, 0) > 0$, $\varrho(i_\sharp + 1, 1) > 0$, respectively.

The sign of $\varrho(i, m)$ determines whether a kernel is *relevant* or *irrelevant*. Roughly speaking, kernels with $\varrho(i, m) \leq 0$ are those whose size does not improve under the scale integration and therefore require stochastic estimates, while kernels with $\varrho(i, m) > 0$ gain powers of the scale parameter and can be controlled deterministically.

In the theorem below we state the *stochastic estimates* for the enhanced noise. Recall that the enhanced noise consists precisely of the finite family of effective force kernels

$$(F_{\kappa,\bullet}^{i,m})_{i \in \{0, \dots, i_b\}, m \in \{0, \dots, 3i\}}.$$

These kernels are the building blocks that encode all stochastic contributions entering the effective equation.

Theorem 9.10. There exist a choice of the counterterms $(c_\kappa^{(i)})_{i \in \{1, \dots, i_\sharp\}}$ in the expression (5.3) for the force F_κ and a random variable $\tilde{R} \in [1, \infty]$ such that $\mathbb{E}\tilde{R}^n < \infty$ for all $n \in \mathbb{N}_+$ and it holds

$$\|K_\mu^{\otimes(1+m)} * F_{\kappa,\mu}^{i,m}\|_{\mathcal{V}^m} \leq \tilde{R} \mu^{\varrho_\varepsilon(i,m)}$$

for all $i \in \{0, \dots, i_b\}$, $m \in \{0, \dots, 3i\}$, $\kappa, \mu \in (0, 1]$.

Remark 9.11. Note that

$$\begin{aligned} \|K_\mu^{\otimes(1+m)} * \dot{F}_{\kappa,\mu}^{i,m}\| &\lesssim \sum_{j=0}^i \sum_{k=0}^m \|\mathbf{B}(\mathbf{P}_\mu^2 \dot{G}_\mu, K_\mu^{\otimes(1+k)} * F_{\kappa,\mu}^{j,1+k}, K_\mu^{\otimes(1+m-k)} * F_{\kappa,\mu}^{i-j,m-k})\|_{\mathcal{V}^m} \\ &\leq \|\mathbf{P}_\mu^2 \dot{G}_\mu\|_{L^1} \sum_{j=0}^i \sum_{k=0}^m \|K_\mu^{\otimes(1+k)} * F_{\kappa,\mu}^{j,1+k}\|_{\mathcal{V}^{1+k}} \|K_\mu^{\otimes(1+m-k)} * F_{\kappa,\mu}^{i-j,m-k}\|_{\mathcal{V}^{m-k}}. \end{aligned}$$

Assuming that the bound from the theorem lemma holds true for all $F_{\kappa,\mu}^{i,m}$ with $i < i_\circ$, or $i = i_\circ$ and $m > m_\circ$ and using

$$\varrho_\varepsilon(j, 1+k) + \varrho_\varepsilon(i-j, m-k) = \varrho_\varepsilon(i, m) - \sigma \quad (9.15)$$

as well as (9.2) one easily shows that

$$\|K_\mu^{\otimes(1+m)} * \dot{F}_{\kappa,\mu}^{i,m}\|_{\mathcal{V}^m} \lesssim \mu^{\varrho_\varepsilon(i,m)-1}$$

for $i = i_\circ$ and $m = m_\circ$. It then follows that

$$\|K_\mu^{\otimes(1+m)} * F_{\kappa,\mu}^{i,m}\|_{\mathcal{V}^m} \leq \int_0^\mu \|K_\eta^{\otimes(1+m)} * \dot{F}_{\kappa,\eta}^{i,m}\|_{\mathcal{V}^m} d\eta \lesssim \int_0^\mu \eta^{\varrho_\varepsilon(i,m)-1} d\eta.$$

Moreover, if $\varrho_\varepsilon(i, m) > 0$ then

$$\int_0^\mu \eta^{\varrho_\varepsilon(i,m)-1} d\eta \lesssim \mu^{\varrho_\varepsilon(i,m)}$$

Hence, if the bound stated in the above theorem is known for the *relevant* effective force kernels, i.e. $F_{\kappa,\mu}^{i,m}$ such that $\varrho(i, m) \leq 0$, it can be easily proved deterministically for the *irrelevant* kernels, i.e. $F_{\kappa,\mu}^{i,m}$ such that $\varrho(i, m) > 0$.

Remark 9.12. Since the kernels $F_{\kappa,\mu}^{i,1}$ with $i \in \{i_\sharp+1, \dots, i_b\}$ are irrelevant no renormalization should be necessary to bound them. This is an intuitive reason why only the counterterms $(c_\kappa^{(i)})_{i \in \{1, \dots, i_\sharp\}}$ are included in the expression for the force (5.3).

Remark 9.13. The bounds for $F_{\kappa,\mu}^{i,m}$ stated in the above theorem and the identity (9.15) imply that

$$\|K_\mu^{\otimes(1+m)} * H_{\kappa,\mu}^{i,m}\|_{\mathcal{V}^m} \lesssim \mu^{\varrho_\varepsilon(i,m)-1},$$

where

$$\langle H_{\kappa,\mu}[\varphi], \psi \otimes \varphi^{\otimes m} \rangle := \sum_{i=i_b+1}^{2i_b} \sum_{m=0}^{3i} \lambda^i \langle H_{\kappa,\mu}^{i,m}, \psi \otimes \varphi^{\otimes m} \rangle.$$

This proves the bound (9.11) with $\beta = \varrho_\varepsilon(i_b+1, 0) > 0$.

9.4 Cumulant estimates

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